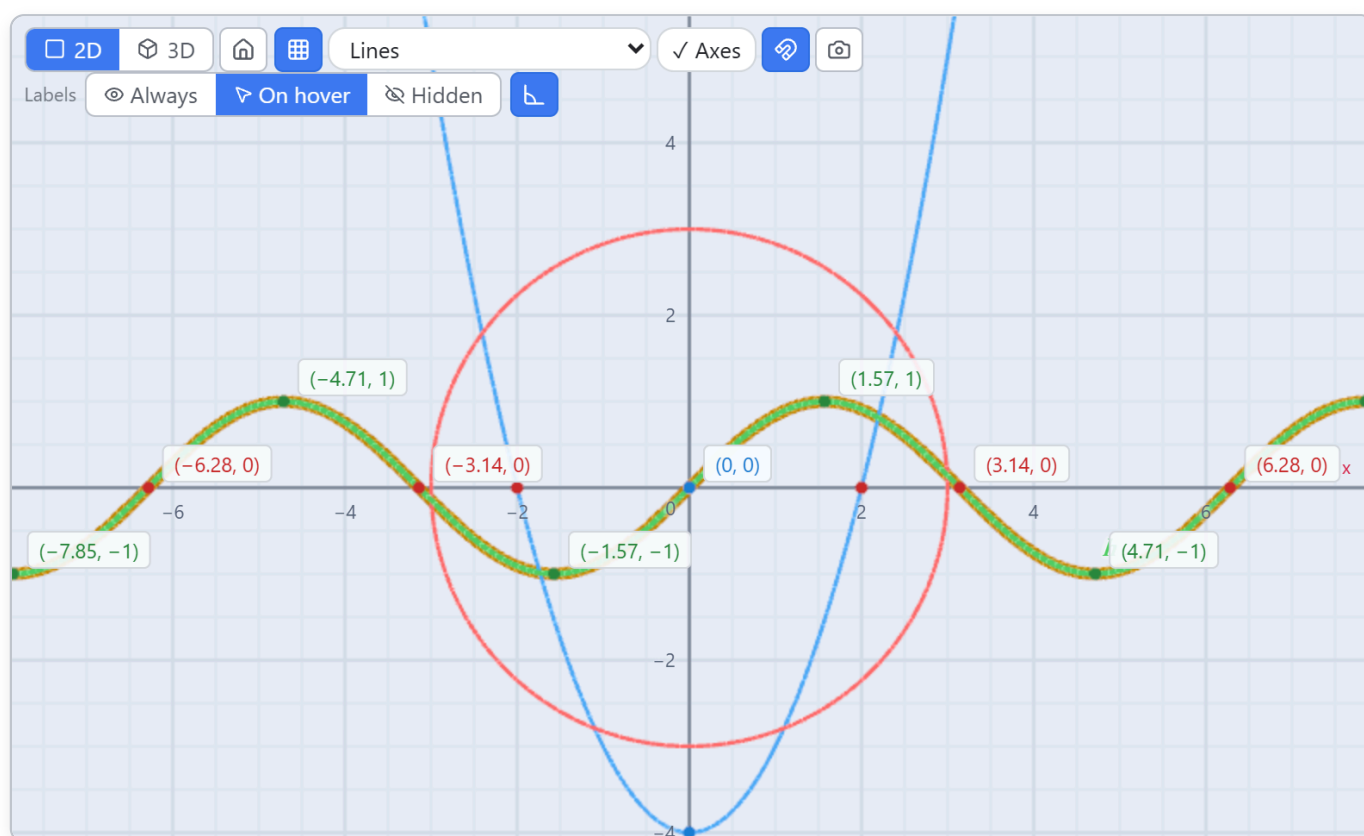
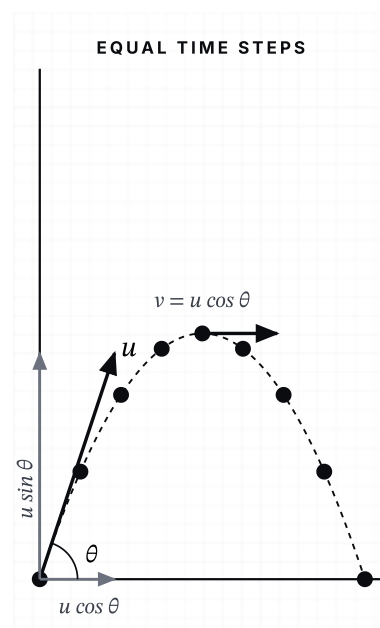


PhysLab

*Maths and physics
you can see.*

A free, fully offline maths and physics engine for **every learner**, from a first look to university, and for the **teachers** who guide them. Type maths the way a textbook prints it. Get the right answer, with its units. See every step. Then watch the idea move.



REAL SCREENSHOT Graphing in PhysLab 0.9, Light theme. $y = x^2 - 4$, $x^2 + y^2 = 9$ and $y = \sin x$ were typed in the command bar; the roots and turning points were found and labelled by PhysLab itself.

10

WORKING MODES,
8 MORE ON THE WAY

2,879

AUTOMATED TESTS,
ALL PASSING

29

REAL-PHYSICS
EXPERIMENTS

0

INTERNET, ACCOUNTS
OR AI NEEDED

WHAT PHYSLAB IS

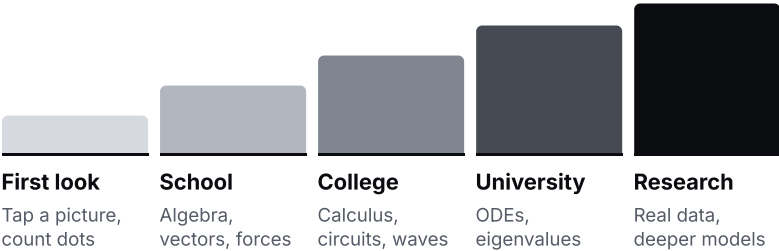
One app. Every step. *Nothing hidden.*

PhysLab is a maths and physics engine you use like a calculator. You type a question the way a textbook prints it. PhysLab gives the right answer with its units, writes out every step a teacher would write, and then lets you **see** the idea: on a graph, in a drawing, or in a real physics simulation you can push and throw.

SIX PROMISES IT NEVER BREAKS

01 Offline and private No internet, no account, no cloud. It works in a village school with no signal.	02 No AI inside Every answer comes from real mathematics and physics that can be checked, never from a guess.
03 Feels like a calculator Natural maths, buttons and short commands. Never anything that looks like programming.	04 The answer comes first Every tool gives the result. The working and the pictures come on top, never instead.
05 Units on every number Each number carries its unit and follows your chosen decimal places or significant figures.	06 Anything solved can be seen A tangent for a derivative, a shaded area for an integral, a triangle for a vector sum.

WHO IT IS FOR: ONE TOOL, FIVE DEPTHS, NOTHING LOCKED



Features are **modes**, like the modes of a scientific calculator, never grade levels. Teachers get their own modes to write questions and read their class's results.

HOW EVERY QUESTION GOES

1

Type it like the book

Fractions, roots and powers look as they are printed.

2

Get the answer

Exact where it can be, with units and your precision.

3

See every step

A plain line and the rule that allowed it.

4

Watch it move

A graph, a drawing or a real simulation.

INSIDE THIS BOOK

01	Calculator & CAS	03
	Textbook maths, exact algebra, every step	
02	Vectors & Geometry	06
	Working the way the book does it	
03	Graphing	09
	Curves, regions, polar, 3-D surfaces	
04	Sandbox	10
	Real physics you can push and throw	
05	Lab Data & GPU Lab	12
	Readings, best-fit lines, a million particles	
06	Teaching	14
	Problem Sets, Question Author, Class Results	
07	Everywhere in the app	16
	Search, lessons, units, files, themes	
08	Progress history	17
	Eleven days from first code to 0.9	
09	The future	20
	Astrophysics, instruments, 3-D and more	
★	The poster	26
	Everything on one page	

UNDER THE HOOD

Electron + React

TypeScript

three.js · WebGPU

Jolt Physics · WebAssembly

SymPy in Pyodide

MathLive

mathjs

GPL-3.0

Coded by Anthropic's state-of-the-art AI, **Claude Opus 5** and **Opus 5.5**, directed by a Class 11 student. No AI inside the app.

63,516	2,879	90	183	11
LINES OF TYPESCRIPT	AUTOMATED TESTS	BUNDLED QUESTIONS	COMMITTS	DAYS TO 0.9

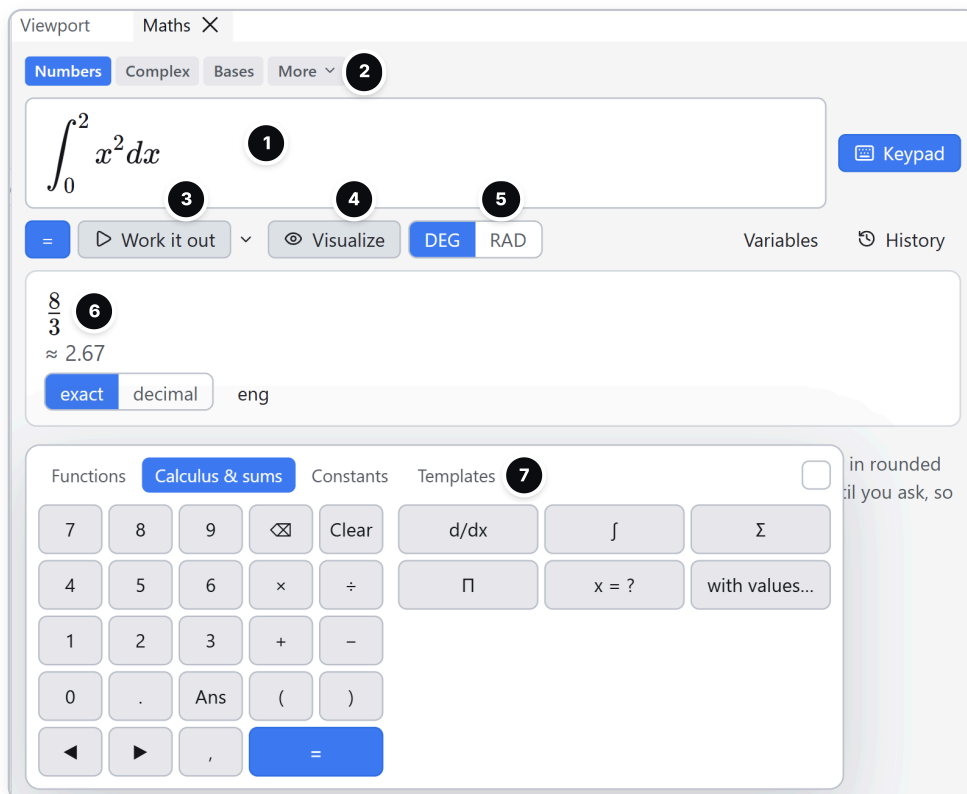
PhysLab · maths and physics you can see

02

THE MATHS SCREEN

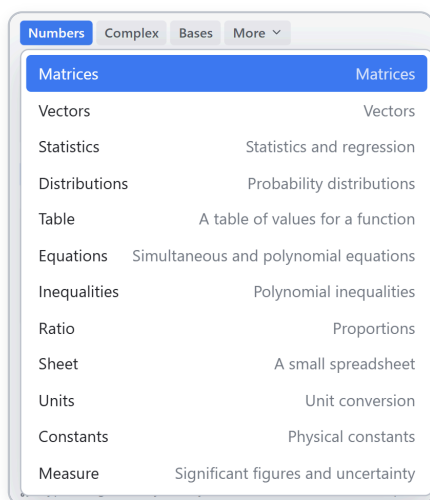
Type it the way the book prints it

One field for every calculation. A fraction, a root, a power or an integral takes shape as you type, the answer arrives first and exact, and *one click draws it*.



- 1 Textbook input.** Typing int, / or ^ gives an integral with limits, a fraction or a power, drawn as the book prints it.
- 2 Modes as chips:** Numbers, Complex, Bases, and More for the rest.
- 3 =** gives the answer; **Work it out** gives the steps, and its arrow picks the kind of question.
- 4 Visualize** draws the calculation: a shaded area, a tangent, a graph.
- 5 DEG or RAD** for sin, cos and tan, one click.
- 6 The answer first, exact:** $\frac{8}{3}$, with ≈ 2.67 in your chosen precision (2 decimal places here).
- 7 The keypad** pops up for keys a keyboard lacks (d/dx, $\int$, Σ , Π , constants). Typing keeps working.

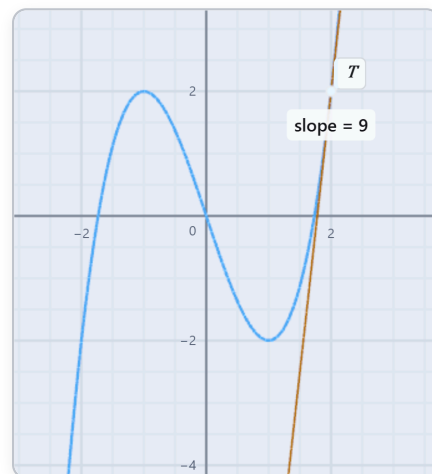
TESTED Typed on the keyboard: int, the limits 2 and 0, then $x^2 dx$, and =. The keypad was opened on its Calculus & sums tab.



TESTED Every fx-991EX mode, in words, plus Units, Constants and Measure. In Matrices, $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = -2$; Units gave 72 km/h = 20 m/s.



TESTED Visualize on $\int_0^2 x^2 dx$: the area under $y = x^2$ from 0 to 2 is shaded, the $\frac{8}{3}$ the screen worked out.



TESTED d/dx of $x^3 - 3x$ at $x = 2$ is 9. Built from the keypad; Visualize draws the tangent at T(2, 2) and writes its slope, 9.

THE ALGEBRA ENGINE

Exact answers, with a teacher's working

Answers stay exact: fractions, surds and i , never drifting decimals. PhysLab checks each worked answer before it shows it, and *says how it checked*.

TESTED Sixteen questions, typed into the 0.9 preview on 25 September 2026, every answer checked again by hand offline · no account

FACTORISE $6x^2 + 7x - 3 = (3x - 1)(2x + 3)$
Split the middle term; multiplied back to check

EXPAND $(2x + 3)(3x - 1) = 6x^2 + 7x - 3$
Each-times-each table; both give 35 at $x = 2$

DIVIDE $\frac{x^3 - 6x^2 + 11x - 6}{x - 1} = x^2 - 5x + 6$
Remainder 0, confirmed by $f(1) = 0$

PARTIAL FRACTIONS $\frac{3x + 5}{(x + 1)(x + 2)} = \frac{2}{x + 1} + \frac{1}{x + 2}$
Cover-up rule; recombined to $3x + 5$

HCF · LCM 12, 18, 30 : HCF = 6, LCM = 180
Prime-factor table; 6 divides each, each divides 180

PRIME FACTORS $360 = 2^3 \times 3^2 \times 5$
Division ladder; 24 divisors in all

COMPLEX $\frac{2 + 3i}{1 - i} = -\frac{1}{2} + \frac{5}{2}i$
By the conjugate; $|z| = \sqrt{13/2}$, arg 101.31°

SOLVE $x^2 + 4x + 13 = 0 \Rightarrow x = -2 \pm 3i$
 $\Delta = -36$; roots add to -4 , multiply to 13

FACTORISE WITH i $x^4 - 16 = (x - 2)(x + 2)(x - 2i)(x + 2i)$
Multiplied back to $x^4 - 16$

DERIVATIVE $\frac{d}{dx}(x^2 \sin x) = 2x \sin x + x^2 \cos x$
Product rule; checked a second way

DEFINITE INTEGRAL $\int_0^2 x^2 dx = \frac{8}{3}$
Power rule, then $F(2) - F(0)$

BY PARTS $\int x e^x dx = (x - 1)e^x + C$
Differentiating the answer gives $x e^x$ back

PROVE $\sec x - \cos x = \sin x \tan x$
Four named steps, each checked at three angles

NORMAL TABLE $P(Z < 1.96) = 0.9750$
Row 1.9, column 0.06; the curve gives 0.975002

MATRICES $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 7 \\ 10 & 15 \end{bmatrix}$
Also $\det = -2$ and the inverse

UNITS $72 \text{ km/h} = 20 \text{ m/s}$
Or type freely: 5 N m to J gives 5 J

Set out the long division

- 7 Write the whole division out as a staircase, with the quotient on top. $(x - 1) \nmid (x^3 - 6x^2 + 11x - 6)$

$$\begin{array}{r} x^3 - 6x^2 + 11x - \\ - x^3 - \\ \hline -5x^2 + 11x - \\ - (-5x^2 + 5x) \\ \hline 6x - \\ - 6x \\ \hline -6 \end{array}$$

quotient $x^2 - 5x + 6$

TESTED Long division as the full staircase, quotient on top, and the rule beside the step.

Multiply by the conjugate

- 1 A complex number on the bottom is not allowed in a final answer, so multiply top and bottom by the conjugate of the bottom. $(a + bi)(a - bi) = a^2 + b^2$

$$\frac{2 + 3i}{1 - i} \times \frac{1 + i}{1 + i}$$

- 2 The bottom becomes a real number, because the i terms cancel. $a^2 - (bi)^2 = a^2 + b^2$

$$(1 - i)(1 + i) = 1 + 1 = 2$$

- 3 Multiply out the top the same way, using $i^2 = -1$.

$$(2 + 3i)(1 + i) = 2 + 2i + 3i + 3i^2 = 2 + 5i - 3 = -1 + 5i$$

TESTED Dividing by a complex number: multiply by the conjugate, with $i^2 = -1$ used where you can see it.

Find each letter

- 4 Because every factor is different and linear, each letter can be read off directly: put in the value of x that makes one bracket zero, and the other terms vanish. cover-up rule

- 5 Put $x = -1$ in, which kills every term except A. $x = -1$

$$3(-1) + 5 = A(-1 + 2) \Rightarrow A = 2$$

- 6 Put $x = -2$ in, which kills every term except B. $x = -2$

$$3(-2) + 5 = B(-2 + 1) \Rightarrow -1 = -B \Rightarrow B = 1$$

TESTED Partial fractions by the cover-up rule: $x = -1$ leaves only A, $x = -2$ only B.

Integrate by parts

- 1 Chose u and dv : u is the factor that gets simpler when it is differentiated. $\int u dv = uv - \int v du$

$$u = x, \quad dv = e^x dx$$

- 2 Differentiated u and integrated dv .

$$du = dx, \quad v = \int e^x dx = e^x$$

- 3 Put them into the formula.

$$\int x e^x dx = x e^x - \int e^x dx$$

TESTED Calculus with every rule named, worked by SymPy running inside PhysLab, offline.

EVERY STEP EXPLAINED

See the working, not just the answer

$x^2 + 4x + 13 = 0$ has no real roots. PhysLab says why in plain English, prints the rule beside every step, finds the complex pair, *then checks it*.

Treat as Solve ▼ Solve a quadratic, real roots or complex ones.

🔍 **Solve** $x^2 + 4x + 13 = 0$ Quadratic formula 📄 Copy

ANSWER

$x_1 = -2 + 3i$ **1**

$x_2 = -2 - 3i$

✓ The two roots are conjugates — that always happens when the coefficients are real — and each one makes the equation zero exactly. **2**

🔍 Let me try first **3** ■ Always show all steps

Read off a, b, c

1 Line the equation up as $ax^2 + bx + c = 0$ and read off the three numbers. $ax^2 + bx + c = 0$

$a = 1, \quad b = 4, \quad c = 13$

Work out the discriminant **5**

2 Square b and take away $4ac$ — the sign of this number decides what kind of roots come out. $\Delta = b^2 - 4ac$

$\Delta = b^2 - 4ac = (4)^2 - 4(1)(13) = -36$

See what kind of root

3 The discriminant is negative. A negative number has no real square root, so the roots are complex.

$\Delta < 0 \Rightarrow$ two complex roots

4 Write the square root of the negative number with i, since $i = \sqrt{-1} \Rightarrow \sqrt{-k} = i\sqrt{k}$ $i^2 = -1$.

$\sqrt{-36} = \sqrt{36 \times (-1)} = 6i$

Use the quadratic formula

5 Put everything into the quadratic formula. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2}$

Read off the roots

6 Take the plus and the minus in turn.

$x = -2 + 3i$ or $x = -2 - 3i$

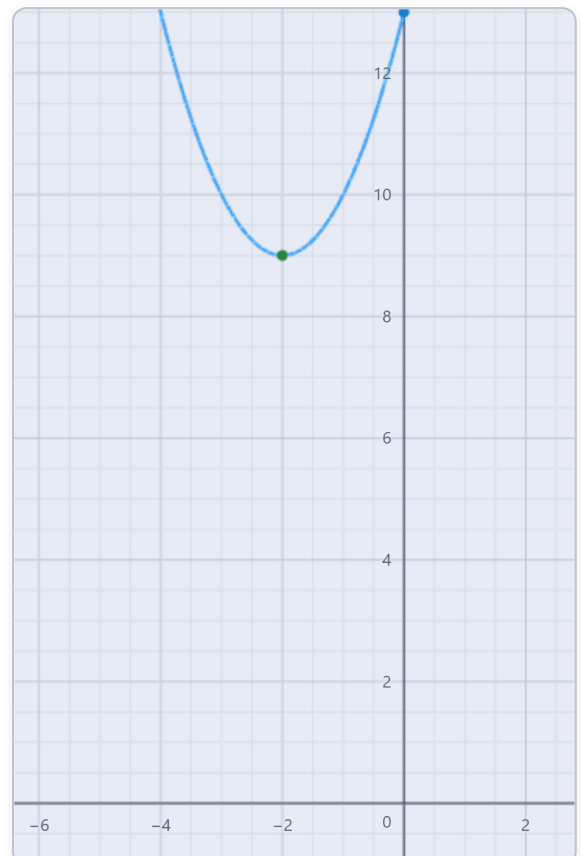
Check against the coefficients **6**

7 Add the roots and multiply them: the sum must be $-b/a$ and the product c/a .

$x_1 + x_2 = -4 = -\frac{b}{a}, \quad x_1 x_2 = 13 = \frac{c}{a}$

📄 Show on the graph **7**

- 1 The answer comes first**, pinned above the steps: $x = -2 \pm 3i$, a conjugate pair.
- 2 The check**: each root makes the equation zero exactly. The tick is that check running, not a promise.
- 3 Try first**. New working arrives with its steps hidden; Show a step reveals one at a time. Let me try first hides them again.
- 4 Plain English** for every step, under a heading that says what it is for: *Work out the discriminant*.
- 5 The rule beside the step**: $\Delta = b^2 - 4ac$, then $\sqrt{-k} = i\sqrt{k}$ and the formula itself.
- 6 A last check** against the coefficients: the roots add to $-b/a$ and multiply to c/a .
- 7 Show on the graph** draws the curve, so the answer can be seen.



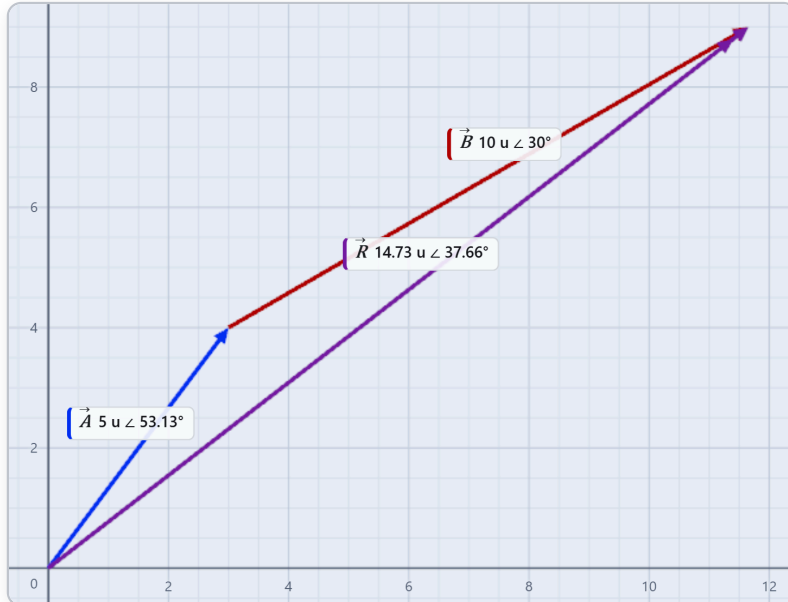
TESTED The proof you can see. The lowest point of $y = x^2 + 4x + 13$ is $(-2, 9)$, above the x-axis, so the curve never crosses it: no real roots, as $\Delta = -36$ said.

TESTED Typed $x^2 + 4x + 13 = 0$, pressed **Work it out**. Treat as: Solve, by the quadratic formula. All seven steps shown.

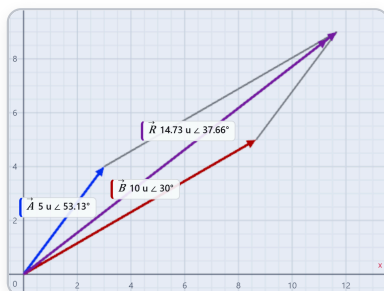
VECTORS

Type a vector like the book

$3\hat{i} + 4\hat{j}$ or $10\angle 30^\circ$: the card reads it at once. One click adds, subtracts or multiplies, the steps are set out the way the book does it, and *Draw on graph* proves the answer.



TESTED Draw on graph, head-to-tail: B starts where A ends and R closes the triangle. Each label gives size and direction: $5 \text{ u } \angle 53.13^\circ$, $10 \text{ u } \angle 30^\circ$, $14.73 \text{ u } \angle 37.66^\circ$.



TESTED Parallelogram, one click: the same $\vec{A} + \vec{B}$.

MORE (9), ONE CLICK AWAY

- Scalar ×
- Unit vector
- Angle between
- Projection
- Equilibrium
- Torque $\vec{\tau} = \vec{r} \times \vec{F}$
- Work $W = \vec{F} \cdot \vec{d}$
- Magnetic force
- Relative velocity

Every operation can show its steps the way the book does, in your precision and angle unit.

- 1 Typed as the book writes it: $3\hat{i} + 4\hat{j}$.
- 2 Or size and angle: $10\angle 30^\circ$.
- 3 Read at once: $\vec{B} = 8.66\hat{i} + 5\hat{j}$, $|\vec{B}| = 10$, $\theta = 30^\circ$.
- 4 Add all: $\vec{R} = 11.66\hat{i} + 9\hat{j}$, $|\vec{R}| = 14.73$, $\theta = 37.66^\circ$, in your precision.
- 5 Draw on graph, laid out head-to-tail, as a parallelogram or from the origin.
- 6 Steps the way the book does it: add the components, then Pythagoras, then $\tan^{-1}$.
- 7 The quadrant is checked before the angle is given.
- 8 Six everyday operations, one click each; More holds nine more.

Vector Calculator

Type like a textbook: $3\hat{i} + 4\hat{j}$, $10\angle 30^\circ$, $\frac{1}{2}\vec{A}$

A

typed

size ∠ angle

graph

$3\hat{i} + 4\hat{j}$ 1

$\vec{A} = 3\hat{i} + 4\hat{j}$ $|\vec{A}| = 5$ $\theta = 53.13^\circ$
A is the first vector: Subtract, Dot and Cross use A and B, in that order.

B

typed

size ∠ angle

graph

$10\angle 30^\circ$ 2

$\vec{B} = 8.66\hat{i} + 5\hat{j}$ $|\vec{B}| = 10$ $\theta = 30^\circ$ 3
Next: Add all gives the resultant of every card; Subtract gives $\vec{A} - \vec{B}$.

+ Add vector

Resultant $\vec{R} = \vec{A} + \vec{B}$

R

$11.66\hat{i} + 9\hat{j}$ 4

$|\vec{R}|$

14.73

θ

37.66°

Draw on graph

5

Steps

Auto

Head-to-tail

Parallelogram

From origin

Next: Draw on graph shows it on the graph, and Steps shows how it was worked out.

1. Write every vector in component form:

$\vec{A} = 3\hat{i} + 4\hat{j}$, $\vec{B} = 8.66\hat{i} + 5\hat{j}$

2. Add the x-components together, the y-components together.

6

3.

$R_x = A_x + B_x = 3 + 8.66 = 11.66$

4.

$R_y = A_y + B_y = 4 + 5 = 9$

5. Resultant vector:

$\vec{R} = 11.66\hat{i} + 9\hat{j}$

6. Magnitude:

$R = \sqrt{R_x^2 + R_y^2} = \sqrt{11.66^2 + 9^2} = \sqrt{216.96} = 14.73$

7. Direction. First the reference angle, ignoring the signs:

$\theta_{ref} = \tan^{-1}\left(\frac{|R_y|}{|R_x|}\right) = \tan^{-1}\left(\frac{9}{11.66}\right) = 37.66^\circ$

8. R_x is + and R_y is +, so the vector is in the 1st quadrant.

7

$\theta = \theta_{ref} = 37.66^\circ$

8

Add all

$\vec{A} + \vec{B} + \dots$

Subtract

$\vec{A} - \vec{B}$

Dot

$\vec{A} \cdot \vec{B}$

Cross

$\vec{A} \times \vec{B}$

Size & angle

$|\vec{A}|$, θ

Components

A_x , A_y

TESTED The Vector Calculator panel after Add all, with Steps open.

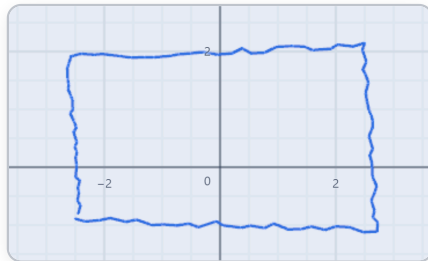
PhysLab · maths and physics you can see

06

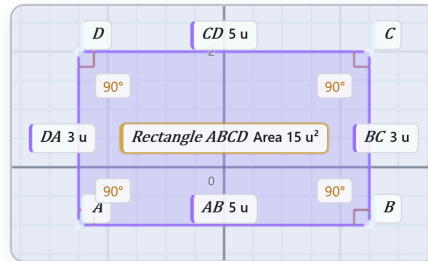
GEOMETRY

Sketch it rough. Get it perfect.

A wobbly loop becomes a true rectangle, triangle or circle, and the Measure panel works out its area the way a textbook sets it out: *formula, values, answer with units.*



TESTED 1. A rough loop, drawn with the Sketch tool.



2. It snaps to rectangle ABCD, 5 u by 3 u, every angle 90°.

Rectangle ABCD ✕ Decompose

AREA

$A = l \times w$ hover to shade

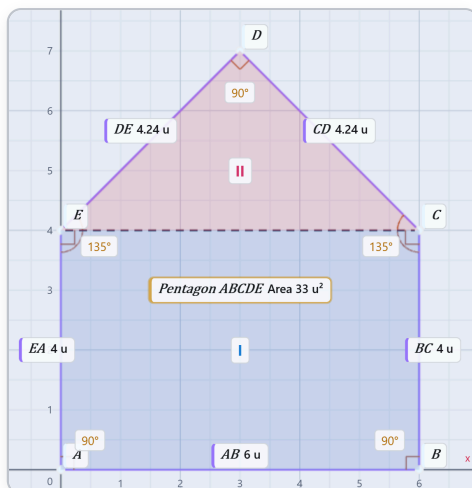
$l = AB = 5 \text{ u}$ $w = BC = 3 \text{ u}$

$A = 5 \times 3$

$A = 15 \text{ u}^2$

3. The area in algebraic form: $A = l \times w$, the values, then $A = 15 \text{ u}^2$.

Decompose · split a shape into ones you know



TESTED The house from the Examples panel, cut along the dashed line.

Pentagon ABCDE ✕ Undo decompose

✕ Break apart

Split into **Rectangles & triangles** Any shape with a formula

Other way 1/4

I Rectangle EABC $A_I = 24 \text{ u}^2$

AREA

$A = l \times w$ hover to shade

$l = EA = 4 \text{ u}$ $w = AB = 6 \text{ u}$

$A = 4 \times 6$

$A = 24 \text{ u}^2$

II Right isosceles triangle DEC $A_{II} = 9 \text{ u}^2$

AREA

$A = \frac{1}{2} \times b \times p$ hover to shade

$b = DE = 3\sqrt{2} \text{ u} \approx 4.24 \text{ u}$ $p = DC = 3\sqrt{2} \text{ u} \approx 4.24 \text{ u}$

$A = \frac{1}{2} \times 4.24 \times 4.24$

$A = 9 \text{ u}^2$

$A = A_I + A_{II} = 24 + 9 = 33 \text{ u}^2$

Each part gets its own formula, then the areas are added: $A = 24 + 9 = 33 \text{ u}^2$.

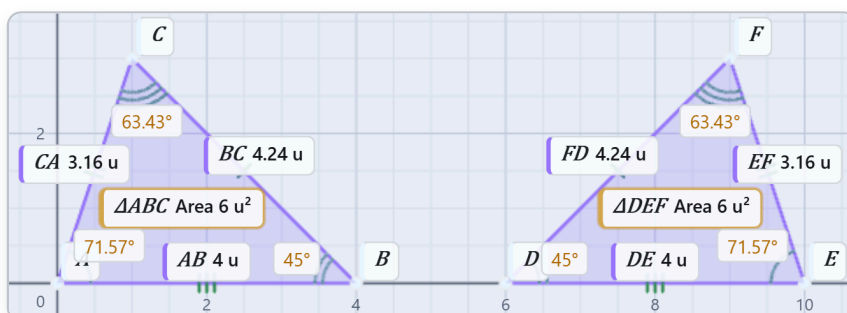
Decompose splits any polygon into rectangles and triangles, adding the corner a cut needs.

Other way shows the other splits: this house has four.

Any shape with a formula allows trapeziums and parallelograms too: fewer pieces, harder formulas.

Hover a formula and its part is shaded on the drawing.

Congruent triangles · with the reason written out



TESTED Shift-click two triangles. Equal sides get matching ticks and equal angles matching arcs, on the drawing itself.

= $\triangle ABC$ and $\triangle DEF$ congruent · SSS

BC · FD $4.24 \text{ u} = 4.24 \text{ u}$

CA · EF $3.16 \text{ u} = 3.16 \text{ u}$

AB · DE $4 \text{ u} = 4 \text{ u}$

$\angle A \cdot \angle E$ $71.57^\circ = 71.57^\circ$

$\angle B \cdot \angle D$ $45^\circ = 45^\circ$

$\angle C \cdot \angle F$ $63.43^\circ = 63.43^\circ$

1. BC = FD = 4.24 u

2. CA = EF = 3.16 u

3. AB = DE = 4 u

4. So $\triangle ABC \cong \triangle EDF$ by SSS (three sides).

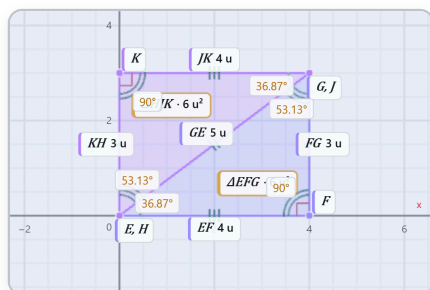
☒ Mark the equal sides and angles on the drawing

All six parts compared, then the rule: $\triangle ABC \cong \triangle EDF$ by SSS. It also knows SAS, ASA, AAS and RHS, and says "similar" when only the angles agree.

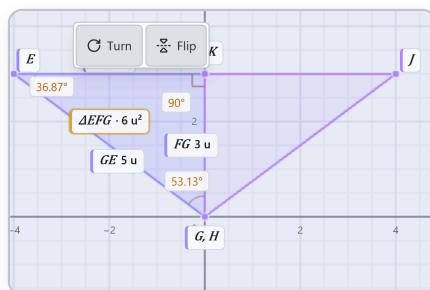
GEOMETRY LEGO, CONSTRUCTIONS, GRID PAPERS

Take shapes apart. Build new ones.

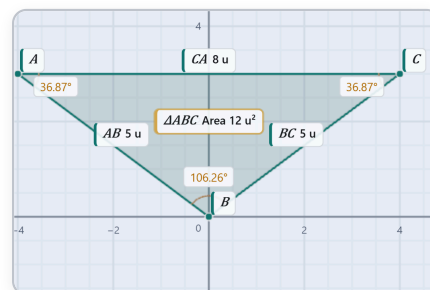
Break apart cuts a rectangle or square along a diagonal and a triangle along a median. Turn and flip the pieces, drag one until it clicks onto a neighbour, and *Fuse* joins them into a new shape.



TESTED Break apart cuts a $4\text{ u} \times 3\text{ u}$ rectangle along a diagonal. Each half is a shape of its own, lettered EFG and HJK.

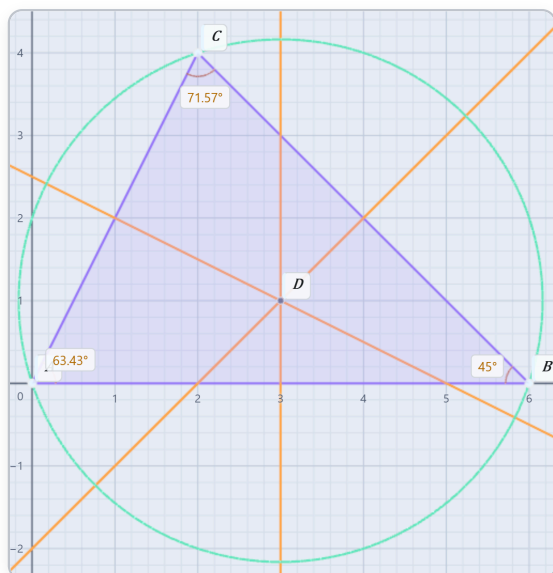


TESTED Flip, Turn 90° twice, then drag one half against the other. PhysLab says "These pieces make a new shape: Isosceles triangle. Press Fuse to join them."



TESTED Fuse: isosceles triangle ABC, sides 5, 5 and 8 u. Its area, 12 u^2 , is the two halves' $6 + 6$.

Constructions · done with tools, not by eye



TESTED Three perpendicular bisectors of triangle ABC meet at one point, D(3, 1), found with Intersect. The circle about D through A passes through B and C too.

ALL MEASUREMENTS		pin = always on drawing
A	(0, 0)	
B	(6, 0)	
C	(2, 4)	
ΔABC	Area 12 u^2	
AB	6 u	
BC	5.66 u	
CA	4.47 u	
D	(3, 1)	
circle, centre D	$r = 3.16\text{ u}$	

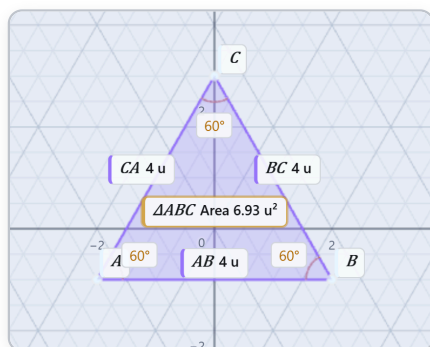
The Measure list reads every object live: D is (3, 1), and the circumcircle has $r = 3.16\text{ u}$, which is $\sqrt{10}$.

The Geometry shelf:

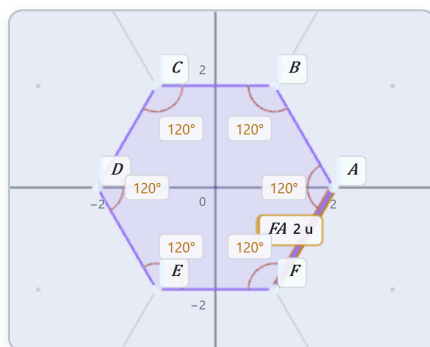
Midpoint, Perpendicular, Parallel, Perpendicular bisector, Angle bisector and Intersect, beside Point, Line, Ray, Segment, Circle, Triangle and Polygon.

Intersect turns a crossing into a named point, D, that the Measure list reads like any other.

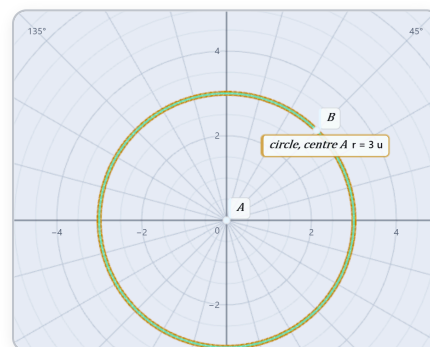
Grid papers · and shapes snap to their points



TESTED Isometric: a sketched triangle snaps to the 60° lattice: equilateral, side 4 u, $A = 4\sqrt{3} \approx 6.93\text{ u}^2$.



TESTED Hexagons: six clicks near the corners give a regular hexagon, side 2 u, $A = 6\sqrt{3} \approx 10.39\text{ u}^2$.



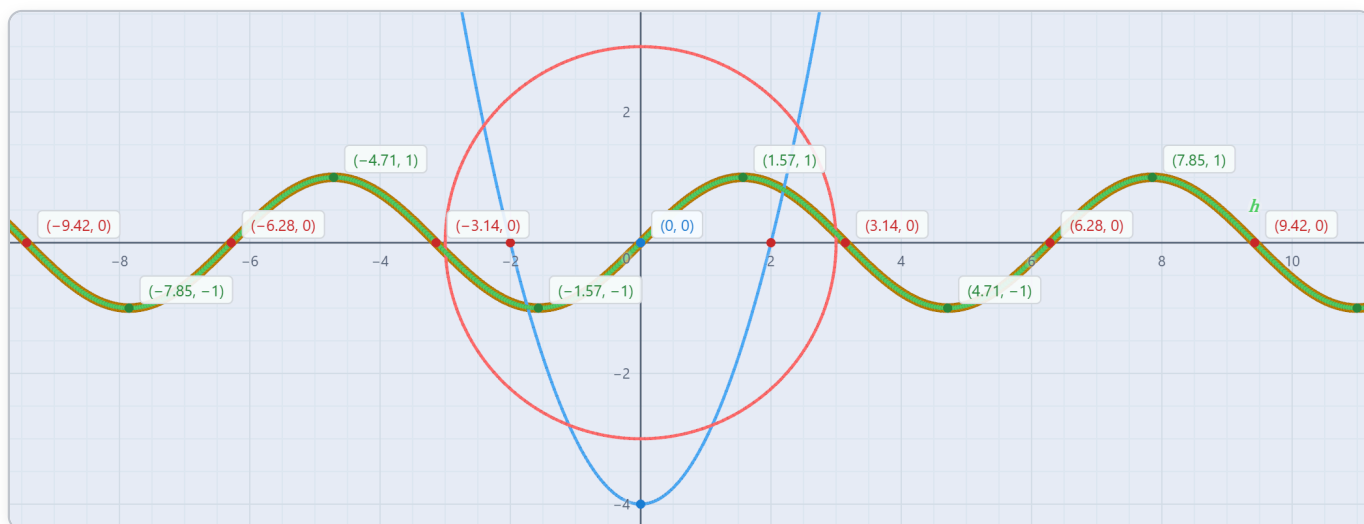
TESTED Polar: rings and angle lines; a circle about the pole snaps to the ring $r = 3\text{ u}$, $A = 9\pi \approx 28.27\text{ u}^2$.

GRAPHING

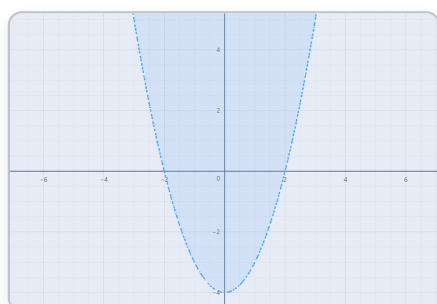
Type the equation. See the curve.

Functions, circles, inequalities, polar roses and 3-D surfaces are typed the way the book prints them, and drawn the moment you press Enter.

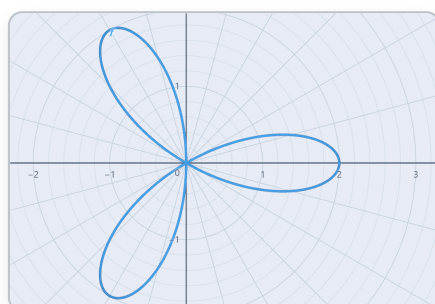
TYPED $y = x^2 - 4$ $x^2 + y^2 = 9$ $y = \sin(x)$ $y > x^2 - 4$ $r = 2\cos(3\theta)$ $z = \sin(x)\cos(y)$ $k = 2$



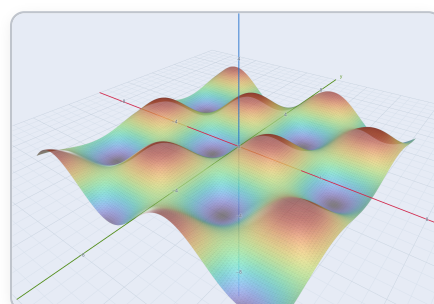
TESTED Three lines typed, three curves on one grid. The sine curve is selected, so every root and turning point is written beside it, such as (3.14, 0) and (1.57, 1), to the chosen two decimal places. The dots on the parabola are its own roots at $x = \pm 2$ and its turning point (0, -4).



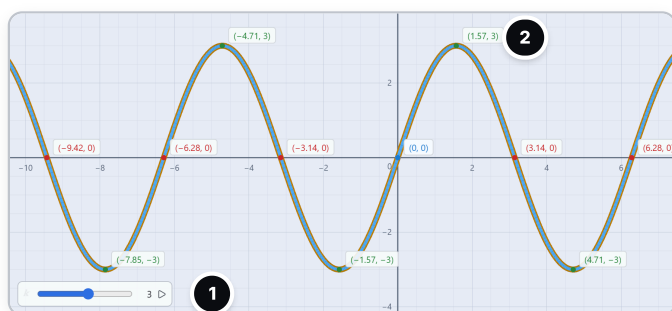
TESTED $y > x^2 - 4$ shades the points above the curve. The dashed edge is not included.



TESTED $r = 2\cos(3\theta)$, a three-petal rose, drawn on polar paper.



TESTED $z = \sin(x)\cos(y)$ in the 3D view, a surface you turn with the mouse.

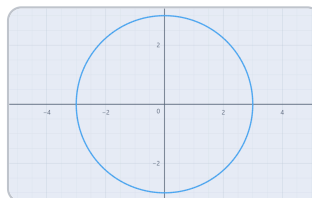


TESTED Sliders. $k = 2$ makes a slider; $y = k\sin(x)$ follows it.

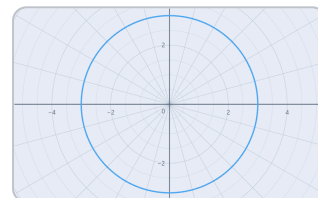
- 1 The slider for k , dragged to 3. Its play button animates it.
- 2 The turning points move with it: the peak is now at (1.57, 3).

Roots and turning points are found for any $y = \dots$ curve and written in the precision you chose in Settings, the same everywhere in PhysLab.

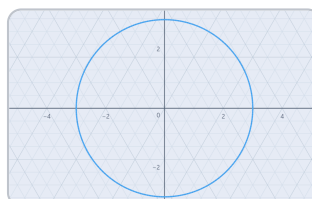
FOUR KINDS OF PAPER, ONE GRID PICKER



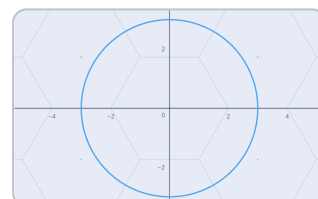
Lines, the everyday square grid



Polar, circles and angles



Isometric, 60° triangles



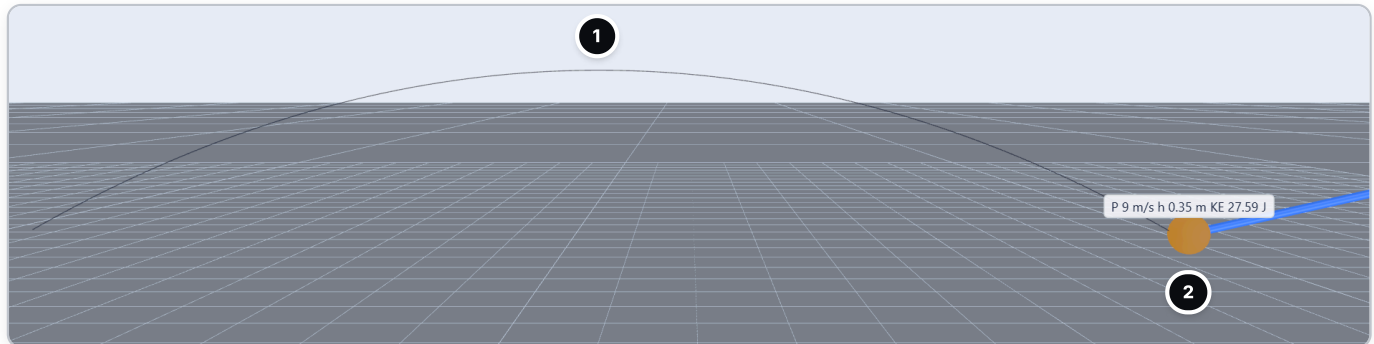
Hexagons, edge to edge

The circle $x^2 + y^2 = 9$ on each paper. Shapes sketched on it snap to that paper's own points.

SANDBOX

Press Play. Then *check* the book.

Balls, crates, ropes and pulleys move by real rigid-body physics from Jolt, the engine behind several big games, and every run gives you numbers to check.



TESTED The Projectile experiment, 14 m/s at 30°, seen from the side. Its card says what the book expects, a range of 17.3 m and a greatest height of 2.5 m, and asks you to read both off the trail.

- 1 The trail it leaves behind: the parabola itself, drawn as it flies.
- 2 Its live label, paused at $t = 1.43$ s just after the bounce: 9 m/s, 0.35 m up, 27.59 J.

Start from an experiment

Q pulley

FORCES

Atwood machine rope over a pulley
1 kg and 2 kg on a rope over a pulley. Both accelerate at $(m_2 - m_1)g / (m_1 + m_2) = 3.27 \text{ m/s}^2$, a third of free fall. Watch m_2 : it falls $\frac{1}{2}at^2 = 0.41 \text{ m}$ in the first 0.5 s, and m_1 rises the same.

Lifting with a pulley rope over a pulley
A 4 kg weight lifts a 3 kg crate off the floor. The pair accelerates at $(4 - 3)g / 7 = 1.4 \text{ m/s}^2$; the rope carries $3 \times (g + a) = 33.6 \text{ N}$. Watch the crate leave the floor at once, and how slowly it gains speed compared with a drop.

Balanced pulley rope over a pulley
Equal 2 kg masses on a rope over a pulley: nothing moves, because the rope pulls both up with the same 19.6 N. Drag one down 1 m and the other rises 1 m — that is all a pulley does: it turns the pull round.

TESTED Search the 29 experiments. “pulley” finds the three that use one. Each card names what joins the objects and states the book’s answer, so the clock can be checked against the formula.

LAUNCHER

m/s at

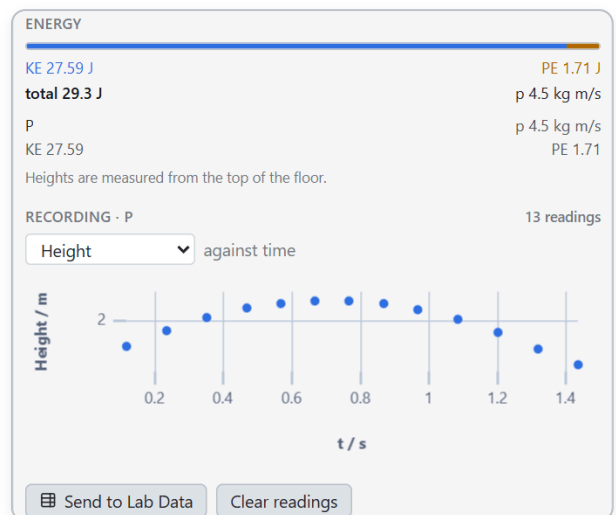
Launch

°

Set

$v_x = 12.12 \text{ m/s} \cdot v_y = 7 \text{ m/s}$

TESTED The launcher does the arithmetic. Type 14 m/s at 30° and it sets $v_x = v \cos \theta = 12.12 \text{ m/s}$ and $v_y = v \sin \theta = 7 \text{ m/s}$, showing both.



TESTED Energy, momentum and a recording, live. The bar splits kinetic from potential energy, with the total and the momentum of everything together. Ten readings a second become a graph, and **Send to Lab Data** turns them into a table (page 05).

Twenty-nine experiments to start from

MOTION 6

Drop a ball
Projectile
Off a cliff
Free fall
Drop on the Moon
Terminal velocity

FORCES 8

Down a slope
Rolling against sliding
Sliding to a stop
Seesaw
Atwood machine
Lifting with a pulley
Balanced pulley
Crane

ENERGY 3

Bouncing ball
Galileo's ramps
Knock it over

MOMENTUM 7

Head-on collision
Sticky collision
Recoil
Into a wall
Newton's cradle
Two trolleys and a rod
Tug on a slack rope

OSCILLATION 5

Pendulum
Rope swing
Bob on two strings
Mass on a spring
Spring on ice

Thirteen join objects with a string, rod, spring, rope, hinge or pulley.

CONNECT TWO OBJECTS

Join anything to anything, and the physics follows

Press **Connect two objects**, click one object, click the other, and pick how they are joined. Each choice is a card that says in one line what it does.

Join A and B with...

String
Pulls when it is taut and never pushes — closer than its length, it hangs slack.

Rod
Keeps them exactly this far apart: it pushes as well as pulls.

Spring
Stretches and squashes with $F = kx$; set k in N/m.

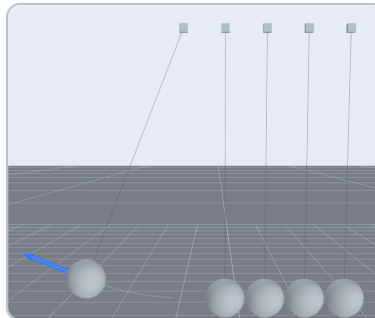
Rope
A real rope: it hangs, bends and goes slack.

Rope over a pulley
The rope runs over a wheel, which must sit above both.
No pulley yet — add one and put it above both A and B.

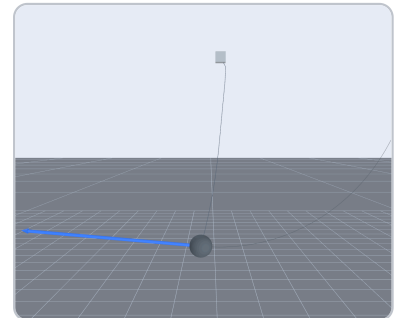
Hinge
Turns about the second object's centre, like a door on its post.

Weld
Glued together: they move as one.

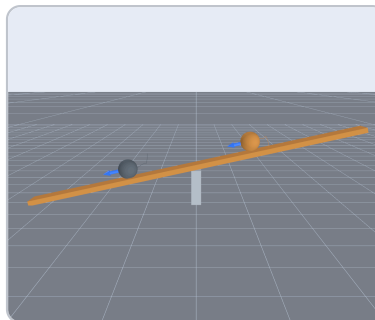
TESTED **Seven ways to join.** String, rod, spring, real rope, rope over a pulley, hinge and weld. The pulley card says what is missing: a wheel above both objects.



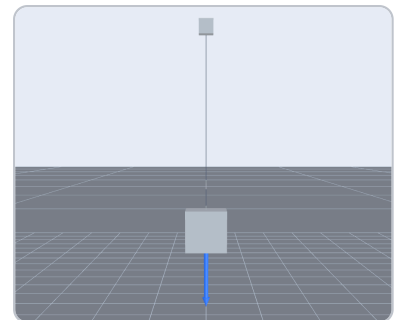
String. Newton's cradle, the first ball just after it swings in.



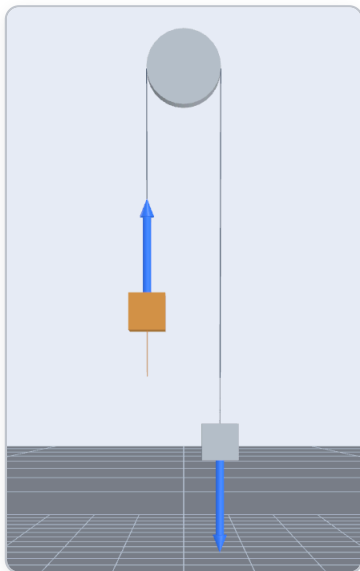
Real rope. A rope swing that bends and goes slack, with its trail.



Hinge. A seesaw, the plank turning on its post.



Spring. A mass on a spring, $F = kx$, bouncing below its hook.



The Atwood machine, checked against the book

1 kg and 2 kg hang on a rope over a pulley. The experiment's card gives the book's answer:

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(2 - 1) \times 9.81}{1 + 2} = 3.27 \text{ m/s}^2$$

TESTED Paused at $t = 0.57 \text{ s}$, the panel shows each block moving at about **1.85 m/s** (momentum 1.85 kg m/s for 1 kg, 3.71 kg m/s for 2 kg). So $a \approx 1.85 \div 0.57 \approx 3.2 \text{ m/s}^2$, the book's value. The lighter block rises as the heavier one falls, each with a velocity arrow.

Total energy read 73.5 J at 0.57 s and 73.43 J at 1.08 s: potential energy turning into kinetic, the total kept.

IS THE PHYSICS REAL?

The automated tests hold the engine to the formulas instead of trusting it. They run with the rest of PhysLab's 2,879 tests, all passing.

- ✓ Free fall covers $\frac{1}{2}gt^2$ in one second
- ✓ Equal masses swap velocities in an elastic head-on hit
- ✓ A sticky hit keeps momentum and loses energy
- ✓ A ball with bounciness e comes back to e^2 of its height
- ✓ A block holds on a gentle slope and slides on a steep one
- ✓ Air drag brings a falling ball to its terminal speed

LAB DATA

Readings in. The *gradient* out.

The table from a practical notebook, with the graph and the write-up attached. Type readings, paste them, import a .csv file, or send them from a Sandbox run.

1 · SANDBOX

Open **Drop on the Moon**, where $g = 1.62 \text{ m/s}^2$. Press Play, then **Send to Lab Data**.



2 · LAB DATA

A new table arrives: 21 readings of t , x , y , v and KE, each column headed with its unit.

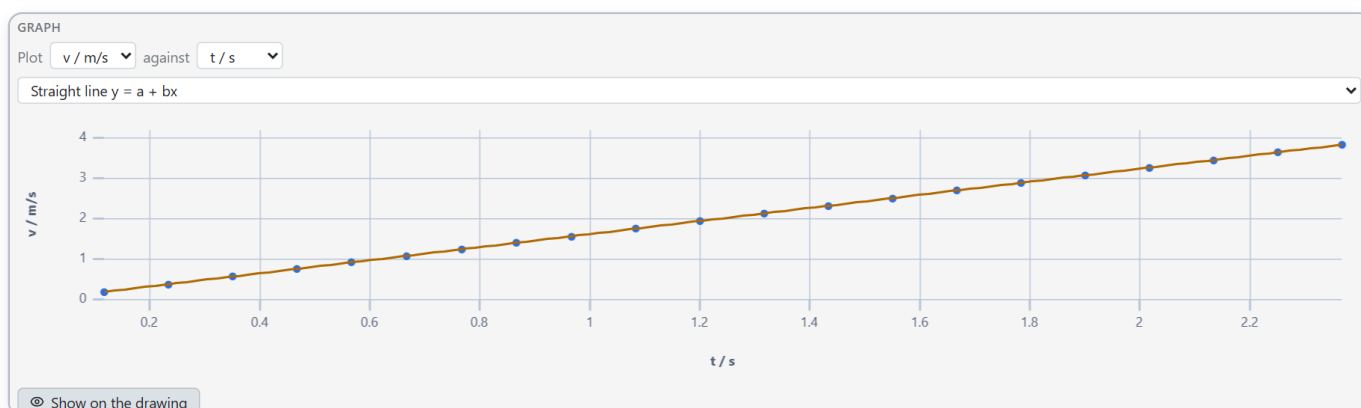


3 · THE RESULT

Plot v against t . The gradient is 1.62 m/s^2 , the Moon's g the run was set up with.

Experiment A — from the Sandb...							
A — from the Sandbox							
t / s	x / m	y / m	v / m/s	KE / J			
1	0.11667	0	5.1882	0.189	0.017861		
2	0.23333	0	5.1543	0.378	0.071442		
3	0.35	0	5.0984	0.567	0.16074		
4	0.46667	0	5.0204	0.756	0.28577		
5	0.56667	0	4.9361	0.918	0.42136		
6	0.66667	0	4.8355	1.08	0.5832		
7	0.76667	0	4.7187	1.242	0.77128		

TESTED The run, as readings. The table is named after the ball it followed, "A — from the Sandbox", and opens in its own tab beside the readings already there. No reading was typed.



TESTED Plot v against t and fit a straight line through all 21 readings. **Show on the drawing** puts the points and the line into the main viewport.

$$v = 1.62 t + 2.69 \times 10^{-6} \quad \textcircled{1}$$

$$r^2 = 1 \text{ — the readings sit very close to this line.} \quad \textcircled{2}$$

$$\text{gradient} = 1.62000 \pm 0.00001 \text{ m/s/s} \quad \textcircled{3}$$

The $\pm$ is how far the line could tilt and still pass through readings this scattered.

Velocity against time: the gradient is the acceleration. $\textcircled{4}$

- $\textcircled{1}$ The equation of the best-fit line.
- $\textcircled{2}$ $r^2 = 1$: the readings sit on the line.
- $\textcircled{3}$ The gradient, 1.62000 ± 0.00001 , with its unit worked out from the column headings.
- $\textcircled{4}$ What the gradient means: the acceleration.

The gradient comes with its meaning. Lab Data knows the common pairs:

d against t	the gradient is the speed
v against t	the gradient is the acceleration
d against t^2	$g = 2 \times \text{gradient}$
T^2 against l	$g = 4\pi^2 \div \text{gradient}$
F against x	the spring constant k

UNCERTAINTY

Error bars and a $\pm$, the way a practical is marked

The example **Free fall: find g from d and t** opens five timed drops, measured to ± 2 cm. A column can be worked out from the others, and a $\pm$ column says how uncertain each reading is.

Free fall					
t / s	1	tsq / s ²	Σ	d / m	$\pm d / m$ 2
		t ²			
1	0.2			0.21	0.02
2	0.4			0.78	0.02
3	0.6			1.8	0.02
4	0.8			3.1	0.02
5	1			4.95	0.02

TESTED 1 The Σ column works tsq out as t^2 and fills itself in for every row. 2 The $\pm$ column gives each distance its uncertainty, 0.02 m.

$$d = 4.92 \text{ tsq} + 0$$

$r^2 = 0.9997$ — the readings sit very close to this line.

$$\text{gradient} = 4.92 \pm 0.04 \text{ m/s}^2$$

From your error bars: the steepest line through them gives 4.98, the shallowest 4.9, and half the difference is the $\pm$.

For free fall the gradient is $g/2$, so $g = 2 \times \text{gradient}$.

TESTED **d against t^2 gives g.** The $\pm$ comes from the steepest and shallowest lines through the error bars, and the value is rounded to the place the $\pm$ supports: $4.92 \pm 0.04 \text{ m/s}^2$, so $g = 9.84 \pm 0.08 \text{ m/s}^2$.

💡 curve fits your readings better — r^2 1 against 0.96.

Use it

TESTED **It says when another shape fits better.** Plotted against t , a straight line scores r^2 0.96 and the curve 1, and one button switches.

CSV in and out. Import reads a .csv file and Export writes one. A block copied from a spreadsheet pastes straight onto the table and can be undone with one click. Tables are saved inside the .phys project.

GPU LAB

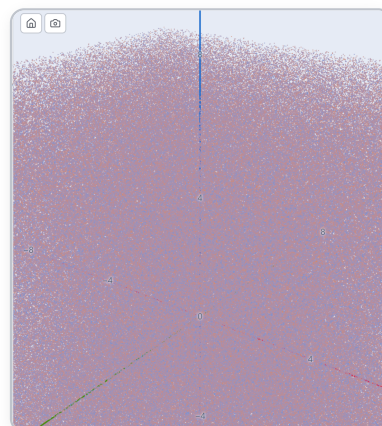
A million charged particles, moved every frame

Each particle feels the force of uniform electric and magnetic fields:

$$\vec{F} = q (\vec{E} + \vec{v} \times \vec{B})$$

With B along z they spiral (cyclotron motion); adding E makes them drift sideways ($\vec{E} \times \vec{B}$ drift). Run 100k, 500k, 1M or 2M particles on the graphics card.

Without WebGPU the processor moves 5,000 to 50,000 instead, never an empty box.



TESTED 1,000,000 particles, each dot one charge.

The graphics card moves the particles (WebGPU).
Renderer: WebGPU · nvidia ampere

Enabled Running Reset

Particles 100k 500k 1M 2M

B field (x, y, z) 0 0 2

E field (x, y, z) 0.3 0 0

q / m 1

TESTED **The panel for that run:** WebGPU on an NVIDIA card, B = (0, 0, 2), E = (0.3, 0, 0), q/m = 1.

PROBLEM SETS

Practice that *names* the slip

Work the question on paper, type what you got, and press Check my answer. A wrong answer is told why it is wrong whenever PhysLab can recognise the mistake.

QUESTION 1 OF 12 Depth 2 · School 0:03

A CAR BRAKES TO A STOP **1**

A car travelling at 30 m/s brakes hard and slows down evenly, coming to rest 9 s later.

YOUR ANSWER

(a) How far does the car travel while it brakes?

m ✗

That is how far it would go at full speed the whole time. Braking evenly, its average speed is only half of that. **2**

(b) Taking the direction it was moving as positive, what is its acceleration while it brakes?

m/s² ✓ **3**

✓ Check my answer ? Next hint

Show the full solution Next >

👁 Show the picture

📖 HINT 1 OF 3 **4**

Steps: every line worked.

Working

1 The speed falls evenly from 30 m/s to 0, so the average speed is halfway between them. $v_{\text{avg}} = \frac{u+v}{2}$

TESTED Question 1 of the Motion set. Part (a) was answered with $v \times t$ on purpose; part (b) was right.

- 1 Fresh numbers every time.** 30 m/s and 9 s here; the next attempt draws new ones.
- 2 The mistake, named.** "That is how far it would go at full speed the whole time. Braking evenly, its average speed is only half of that."
- 3 A right part is ticked,** -3.333 m/s^2 , marked within the author's tolerance.
- 4 Hint gives one step, not the answer:** hint 1 of 3, with the rule beside it.

Worked steps that fade as you improve

Every line
the whole working



Some lines hidden
you fill them in



Headings only
you write the lines

Two right in a row shows less; a slip shows more again.

Ninety questions ship with PhysLab

First look	12
School maths from Numbas	22
Physics mechanics	20
Physics fields	20
University	10
Sample	6

Offered all together, or as 22 topic sets **90**

76 of them carry a picture, 10 a motion graph and one a live Sandbox experiment, shown with the student's own numbers. A teacher's own .pqjson set or a Numbas .exam file opens beside them.

📁 Open a question file...

DEPTH

Every depth Depth 1 · First look Depth 2 · School Depth 3 · Pre-university **Depth 4 · University**

Depth 5 · Research

🌟 Start "Every sample question" — 8 questions

TESTED The Depth selector is a filter, never a lock. From Depth 1 · First look to Depth 5 · Research; picking Depth 4 · University starts the 8 questions tagged with it.

Save my results for my teacher

A file with each part you answered, whether it was right first time and in the end, your marks, the hints you used and the time you took — never sent anywhere by PhysLab. You choose who it goes to.

Student A

Save my results for my teacher

Saved. Give this file to your teacher — PhysLab never sends it anywhere on its own.

TESTED Save my results for my teacher. A file the student hands in; PhysLab never sends it anywhere.

QUESTION AUTHOR

Write a question once. Every student gets their own numbers.

QUESTION AUTHOR

Your name — every question you start is yours under CC BY 4.0

A. Teacher

THIS SET

1. A braking train

+ New question Open a question file...

Save the set for Problem Sets

Save as a Numbas exam

QUESTION 1

Title

A braking train

Licence CC BY 4.0

Credit to A. Teacher

Anyone may use and change it, as long as they name you.

TESTED A new question, credited to its author under CC BY 4.0, ready to save for Problem Sets or as a Numbas exam.

VARIABLES

Give each variable a range, a list or a formula. A condition can drop variants that do not work, and a preview shows ten students' numbers with every answer worked out.

QUESTION & SCENE

The words and the maths, plus something to see: a picture in Graphing (a curve, a curve in pieces, an area between, a tangent), a motion drawn as graphs, or an experiment set up in the Sandbox.

SOLUTION

Each part is a number, a formula, a choice, a vector, a matrix, a set of roots, a function or a proof, with its unit, its tolerance and its marks. Common mistakes get their own "why". The working can show every line, some lines or headings only, and PhysLab can generate the steps again for every student's numbers.

Credit travels with the question. Every question starts under CC BY 4.0 in the author's name (CC BY-SA 4.0 and CC0 1.0 can be chosen instead). One brought in from a file keeps its own author's licence and credit.

974

NUMBAS EXAMS
GATHERED

6,684

LOOSE QUESTIONS
GATHERED

272

PASSED THE
LICENCE CHECK

22

HAND-CHECKED, IN
THE APP

A Numbas .exam file opens in Problem Sets and in the Author, which says in plain sentences what it had to leave out; a PhysLab set saves back out to .exam.

CLASS RESULTS

See which question parts worked, on your own PC

Open the result files the class saved from Problem Sets. PhysLab works out three classic numbers for every question part on this computer and sends nothing anywhere. Weak parts get a plain sentence, and the table saves as CSV.

Question	Part	Attempted	Facility	r_{pb}	D27
Two resistors side by side	1	8	38%	0.68	0.75
Two resistors side by side	2	8	38%	0.68	0.75
Heat from a current-carrying resistor	1	8	50%	-0.15	0
Does not separate strong and weak answers.					
A potential divider	1	8	50%	0.47	0.75
Current and power in a resistor	2	8	63%	0.7	1
Two resistors in series	2	8	63%	0.7	1
Two resistors in series	1	8	88%	0.39	0.5
Current and power in a resistor	1	8	100%	—	0
Nearly everyone got it right first time — too easy to tell strong and weak students apart.					

TESTED Eight sample result files for the Circuits set, saved from Problem Sets for this preview by a script that answered to a fixed pattern. The hardest parts come first.

Facility

The share of the class right first time.

 r_{pb} , corrected correlation

Do students strong on the rest of the test do well here? 0.2 or more is good.

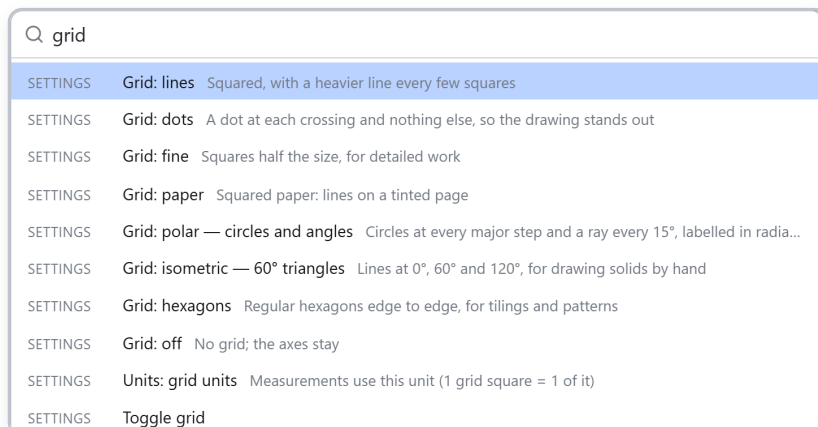
D27, discrimination

The top 27 % of the class minus the bottom 27 %. 0.2 or more is good.

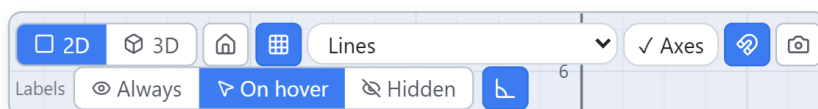
EVERYWHERE IN THE APP

Find anything. *Keep everything.*

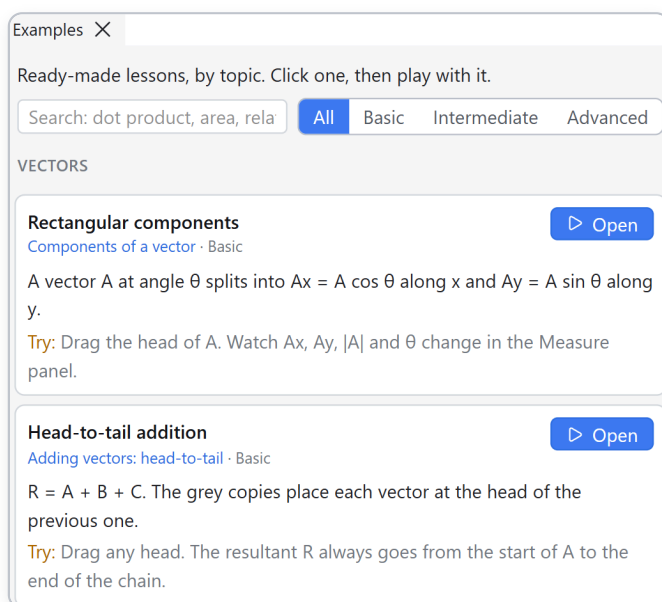
Every mode shares one search, one settings menu, one kind of file and one set of shortcuts. All of it runs offline, with no account, and nothing is sent anywhere.



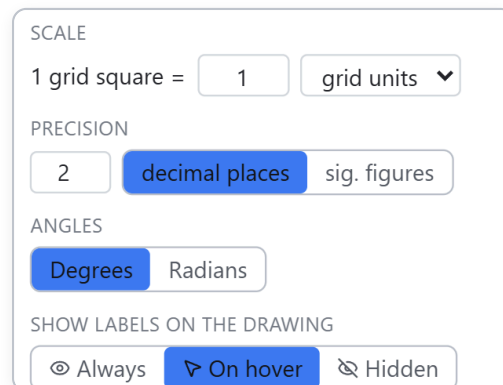
TESTED **Ctrl+K searches everything.** "grid" finds every grid pattern and every unit setting; "area" finds the lessons and the Geometry mode.



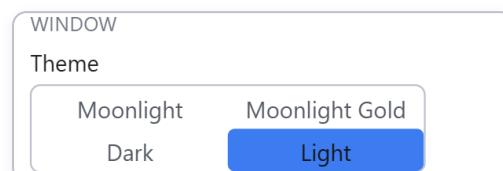
TESTED **The drawing's own bar.** Left to right: 2D or 3D view, reset the view, the grid and its picker, axes, snapping, and **save the drawing as a PNG**. Below: labels **always**, **on hover** or **hidden**, and angle marks.



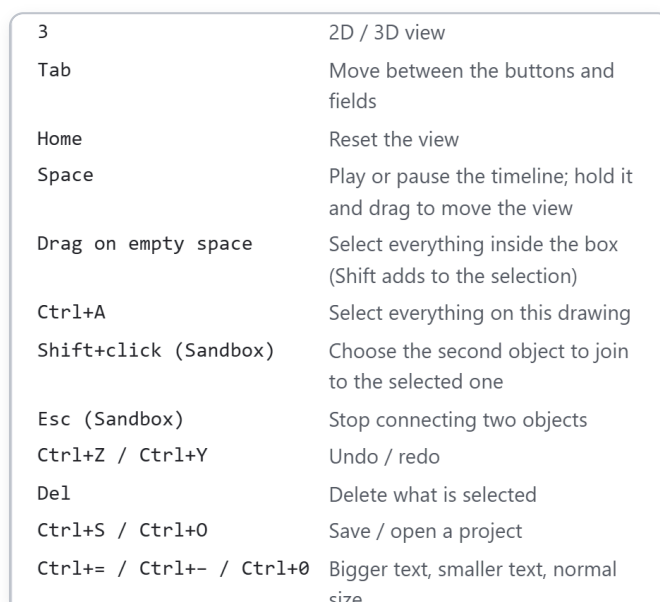
TESTED **Examples: 20 ready-made lessons**, grouped by topic and tagged Basic, Intermediate or Advanced instead of a grade or a chapter. Open one and it builds the lesson for you.



TESTED **Units and precision for the whole app.** A grid square can be a unit, mm, cm, m, km, inch or foot; numbers follow your decimal places or significant figures, in degrees or radians.



TESTED **Four themes:** Moonlight, the default, Moonlight Gold, Dark, and Light, the one printed here.



TESTED **Help ▶ Keyboard and mouse** lists every shortcut, from 3 for the 2D or 3D view to Ctrl+S to save and Ctrl+= for bigger text.

FILES YOU KEEP

Ctrl+S saves the drawing, the Sandbox and every Lab Data table in one .phys file. Unsaved work is copied aside every minute.

OFFLINE, NO ACCOUNT

Everything runs on the computer in front of you: no sign-in, no internet, no cloud, and no AI inside.

MADE FOR FINGERS TOO

Every button in Problem Sets, the Question Author and Class Results is at least 44 px tall, measured in this preview.

TEXT YOUR SIZE

Ctrl+= and Ctrl+- make every word bigger or smaller, and the app remembers.

THE WHOLE STORY

Eleven days, from nothing to 0.9

PhysLab's first line of code was written on 14 September 2026. What follows is every version since, in order — not the code, but what a learner gained.

- 14–15 Sep

●

0.1.0 FIRST WORKING APP

The first six modes arrive — draw and measure vectors and shapes, plot a typed equation, and calculate in natural textbook maths, inside one dockable window.
- 15 Sep

●

Plan v3 POLISH THAT MADE IT USABLE

Drawing stops fighting back: a right-click, Enter or double-click finishes a shape, Backspace undoes the last point, and a guided tour explains the rest.
- 16 Sep

●

0.2.0 PROBLEM SETS

PhysLab starts asking questions back — ten topics generate practice problems, hints reveal one step at a time, and the checker names the actual mistake instead of just marking it wrong.
- 16 Sep

●

0.2.1–0.2.2 THE BLACK VIEWPORT

The most instructive bug in the app's history: the screen opened black on first run. It's fixed — and the light theme becomes properly usable for the first time.
- 17 Sep

●

0.2.5 LAB DATA

Real experiment readings become a graph with a meaning — a fitted line, its gradient in real units, and uncertainty bars that can actually change the answer.
- 18 Sep

●

0.3.0 THE SANDBOX REBUILT

Physics stops cheating: mass finally matters when you drag something, balls stop rolling forever, and editing one object no longer resets the whole scene.
- 19 Sep

●

0.3.2 PURE MATH

The calculator shows its full working for the first time: exact fractions, one plain-English line per step, every step checked before it's shown.
- 20–21 Sep

●

0.3.4–0.3.10 SEVEN RELEASES, TWO DAYS

The Sandbox becomes trustworthy — lifting, dragging and resetting all work, and ropes and pulleys hang and swing properly. Geometry learns congruent triangles, and the whole app settles into one calm, consistent look.
- 21 Sep

●

0.5.0 TESTS AS BOUNDARIES

Nothing new to see, but everything is safer: files, the command bar and every colour are now tested at the exact joins where earlier bugs had hidden.
- 21 Sep

●

0.6.0 MOONLIGHT

A new theme easy on tired eyes, on by default; one natural maths screen instead of a fake handheld calculator; and Sandbox numbers that now match the textbook formula.
- 22 Sep

●

0.6.1 SMALL FIXES, REAL ONES

The Vector panel stops forcing two vectors on you before you've drawn any, deleting a shape no longer leaves its sides behind, and a warm gold theme joins Moonlight.
- 23 Sep

●

0.7.0 RELEASED

Shapes can be broken into pieces and snapped back together, three new grid patterns arrive, and a teacher can bring a whole class's questions in — including from Numbas.
- 25 Sep

●

0.9 preview WHERE THIS BOOK STANDS

The biggest jump yet: teachers can author vector, matrix and equation answers, every question earns its own picture, students can send results home, and 2,879 tests stand behind all of it.

THE PLAN

Eight waves, two landed.

0.9 is a program of eight waves, each reviewed before the next begins. Waves 1 and 2 are merged — Wave 2 grew in the doing, picking up a results dashboard and forty new bundled questions.

THE EIGHT-WAVE ROADMAP

Wave 01

Foundations

Lego, vectors, working and 3-D turning fixed; the release gate built.

LANDED

Wave 05

Lab Data grows up

A graph builder, triggered capture, real lab apparatus on screen.

AHEAD

Wave 02

Finish the question engine

Calculus/trig steps, exact grid numbers, new answer kinds, Numbas, a picture per question.

LANDED

Wave 06

Numbers and statistics

Complex numbers, a Sandbox Pro, a statistics mode, a gentler first run.

AHEAD

Wave 03

Astrophysics

A scale ladder from Earth to the observable universe; orbits, stars, galaxies.

AHEAD

Wave 07

Sharing and polish

A visual redesign, a calmer interface, touch-friendly controls.

AHEAD

Wave 04

Formulas and 3-D

A searchable formula library, graph sound, solid shapes with a build animation.

AHEAD

Wave 08

Prove it, then speed it up

Property tests, four-theme screenshots, one measured optimisation pass.

AHEAD

WHAT ACTUALLY SHIPPED, GROUPED BY WHAT IT GAVE YOU

Thirty-six tracks have merged into the 0.9 build so far, each with its own tests, typecheck and lint run clean before the next — grouped here by who felt the difference.

TRUSTED AGAIN, FROM THE GROUND UP

LEGO	Breaking a shape apart now leaves clean, lettered pieces that fuse back correctly.
VEC	A vector's card and its arrow are one thing — delete, undo or rename either, and both follow.
STEPS	Every worked line is now true exactly as written, set out the way a textbook sets it out.
AUTH	Question Author is born: teachers write questions, check ten random variants, and export them to share.
VIEW3D	Every 3-D view turns properly by mouse, keys or touch, and GPU Lab now works even without WebGPU.
SEG	A drawn segment measures exactly what it shows, and the toolbar behaves the same at every window size.
GATE	One command checks the tests, the types, the lint and both builds before anything can ship.
INT	The Sandbox panel holds still, textbook names and equations appear throughout, practice marks fairly.

THE CALCULATOR LEARNS NEW TRICKS

QT	Type prove(...) and watch a trig identity proved line by line — a false one is refused, not excused.
QK	Axis numbers now sit exactly on the grid lines and always include zero, on paper and in 3-D.
QCa	Integrate and Differentiate get real textbook steps — power, parts, substitution — each one checked true.
QCb	Those calculus steps now show up everywhere: in Working, from the command bar, and inside a question.
QZ	Type $P(Z < 1.96)$ or $\text{invnorm}(0.975)$ and get the printed-table answer, with a calculator check beside it.
QG	Every grid line is now exactly one sharp pixel wide, at any zoom or pan, in all four themes.
QA	A field of hundreds of vectors now turns and zooms smoothly — drawn as one batch instead of hundreds.

EVERY QUESTION PROVES ITSELF

QV1	Every question now has a picture — a graph, a flight, a shaded area — drawn from its own numbers.
QV2	One "Show the picture" button opens it, greyed out until the question is actually answered.
INT	Integration fix: a picture, a motion graph or a revealed answer could still give the game away early.

TEACHERS CAN AUTHOR ALMOST ANY QUESTION

QE1	Question files can hold a vector, matrix or root-set answer, or a measured value with its uncertainty.
QE2	A function answer is checked in its own equation, and a wrong part (a) no longer costs part (b).
QE3	A teacher's Numbas questions — matrices, adaptive marking, conditions — now import and mark the same way.
QE4	Students can answer all these new kinds in Problem Sets, filter by depth, and press Go deeper.
QE5	Question Author gets an editor for each new kind, plus conditions that keep only sensible random variants.

RESULTS TRAVEL FROM STUDENT TO TEACHER

QR1	A student can save their results to a file for their teacher — opt-in, and never sent anywhere.
QR2	A teacher opens a class's result files in Class Results and sees each part's difficulty at a glance.

MORE TO PRACTISE

QP1	20 new physics questions on motion, forces, energy and momentum, each with its own picture and steps.
QP2	20 more, on circuits, gases, waves and rotation.
QN1	All 974 exams and 6,684 questions gathered from Numbas; 272 passed the licence check.
QN2	A curated Numbas school-maths set joins the bundle, credited and licence-checked.
QD1	A "first look" set for a first-time learner — tap a picture, count the dots.
QD2	Ten university-level questions — ODEs, eigenvalues, normal modes — each checked by hand.
QL	A Lego-fill question: cut the pieces, rebuild the shape; marked right anywhere, turned or flipped.

FOUND BY THE CREW, FIXED BEFORE ANYONE SAW THEM

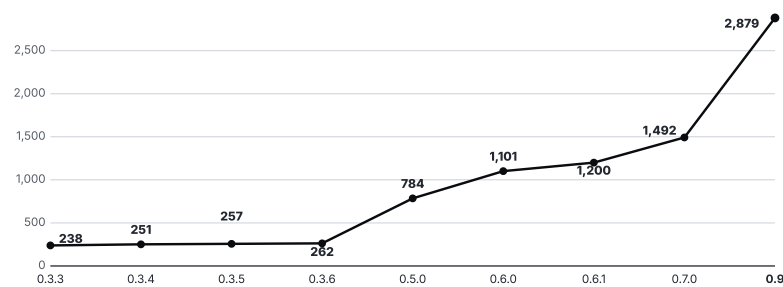
GFQ	Eleven marking bugs closed, from a zero answer to a damaged question file, and more.
GFV	The vector calculator stops losing a student's cards on File → New, Open or Example.
GF5	Worked steps keep their correct signs, and never call a false trig identity "nearly right."
GFR	The 3-D camera, the Sandbox clock and the grid's origin all settle their last rough edges.

VERIFICATION

How we know it works

Nothing here is a promise on faith. Every version added tests before it added features, every change was read by an independent reviewer before it merged, and the numbers below are the ones the project's own history file records.

TESTS, FROM THE FIRST REPAIRS TO TODAY



Source: each release's own test count in HISTORY.md; the 0.9 figure is the full suite run at branch head 382b5be.

183

COMMITTS
146 at 0.7.0

63,516

LINES OF TYPESCRIPT
43,596 at 0.7.0

110

TEST FILES
60 at 0.7.0

THE QUALITY GATE

One command has stood between any change and the next since Wave 1: it runs the whole test suite, checks every type, lints every file, and builds the app twice. Nothing merges until all four are green.

Tests → Types → Lint → Both builds

Physical constants sit at CODATA 2022; a written accuracy policy fixes the marking tolerance at 2%, stated once for the whole app.

SIX RULES, UNBROKEN SINCE DAY ONE

- **No AI inside** — fully offline, no accounts, no cloud.
- **Feels like a calculator**, never like programming.
- **The answer comes first**; the working is a bonus.
- **Every number carries** its unit and your precision.
- **Features are modes**, never grade levels.
- **Anything solved** can be seen.

BUGS WORTH REMEMBERING

The Sandbox's own rebuild began because the student who uses this app said so plainly: *it should be the student's most favourite feature*. It took three real bugs — two of them left as standing lessons in the project's own engineering notes today.

0.2.1 → 0.2.2

The screen that opened black

On first open, the viewport showed nothing but the letters x, y and 0. The cause: the very first frame drew with a stand-in camera at the wrong zoom, and nothing guaranteed a second, correct frame after the real one arrived. The fix watches the camera itself change.

0.3.0 · JOLT CONSTRAINTS

The rope that held a ball in mid-air

A rope should swing. Instead, a slack string once held a ball perfectly still, half a metre below its pivot — the constraint had been given the bodies' world positions, which pins them where they stood. It now attaches at [0,0,0] in each body's own frame, never in world space.

0.3.0 · JOLT PHYSICS

The shape that vanished, silently

Jolt Physics owns its shapes by reference count. Free the settings object a moment too early, and the body's shape disappears with it — no error, no warning. It is now a standing rule in the project's own notes: never free a ShapeSettings before Create() has used it.

CHECKED BY REVIEWERS WITH NO STAKE IN THE CODE

PhysLab is reviewed while it is being built, not after. Independent AI systems, with no memory of writing the code and nothing to gain from calling it finished, read every change looking for what its own author would miss. Historically, about one third of what they report turns out to be real once it is checked line by line against the current source; the rest describes code that had already changed, or was simply misread — one suggested "fix" would have crashed every ramp in the Sandbox. Every finding is verified before anything is touched. None of this reaches the product: the app a student installs has no AI inside it at all, exactly as rule one has said since the very first day.

AFTER 0.9

09

Where PhysLab goes *next*

Seventeen ideas are on the list. Each was researched, checked against the code and designed before a line of it is built. These pages draw them as they are meant to work: illustrations of the design, never screenshots.

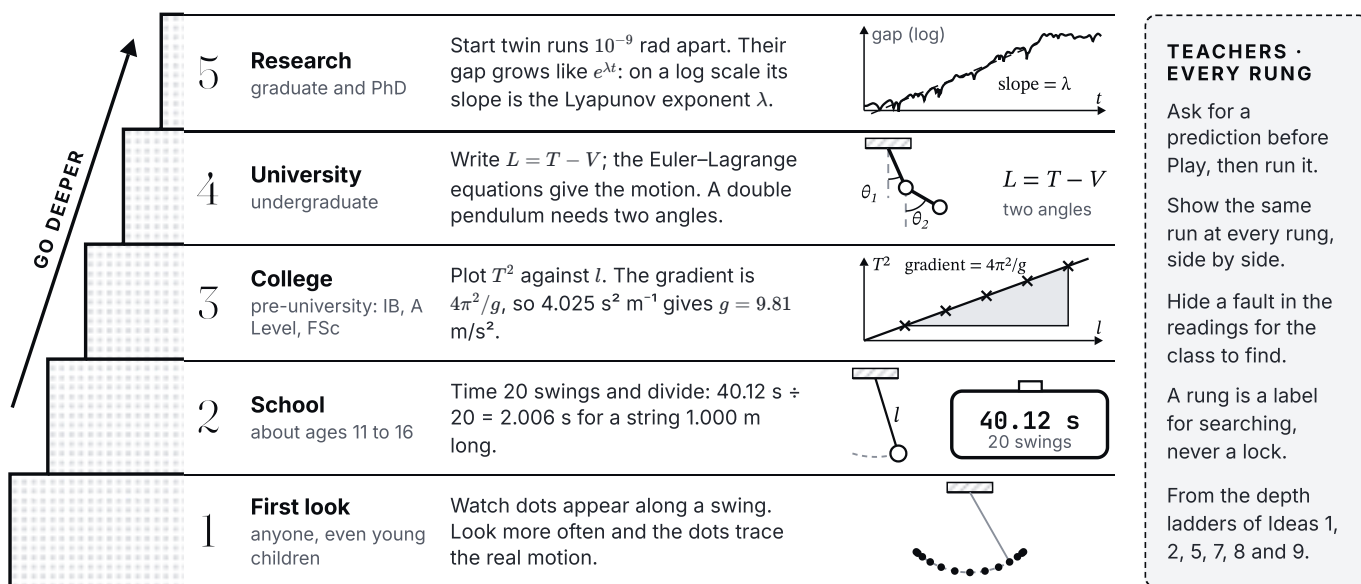
IN 0.9 NOW **Idea 1: question banks, Geometry Lego and three new grids.** Built and tested in this preview: 90 bundled questions, Numbas import and export, and polar, isometric and hexagonal paper.

FUTURE **Everything else in this section.** Researched and designed, not built yet; every number drawn here was worked out and checked. Phone apps, Android first and then iOS, come after Windows.

THE DEPTH LADDER

One tool, five depths. Nothing is locked.

Every tool opens at a first look. **Go deeper** adds a layer on the same screen, right up to research. There are no grade gates: anyone may climb. Here is one pendulum, all the way up.



THE MAP

Eight modes are waiting

PhysLab already keeps a place for eight more modes. Each has a plan that runs from a first look to university and beyond. Astrophysics would be a new, ninth mode.

Instruments · a vernier, a screw gauge and ten more [Idea 2](#)

Mechanics · a push on a block, up to Lagrangian mechanics
[Idea 7 · roadmap 18](#)

Proofs · a proof builder with no programming [roadmap 35](#)

Waves & Sound · hear a graph; play what is simulated
[Idea 3 · roadmap 19](#)

Optics · rays and lenses, up to interference [roadmap 20](#)

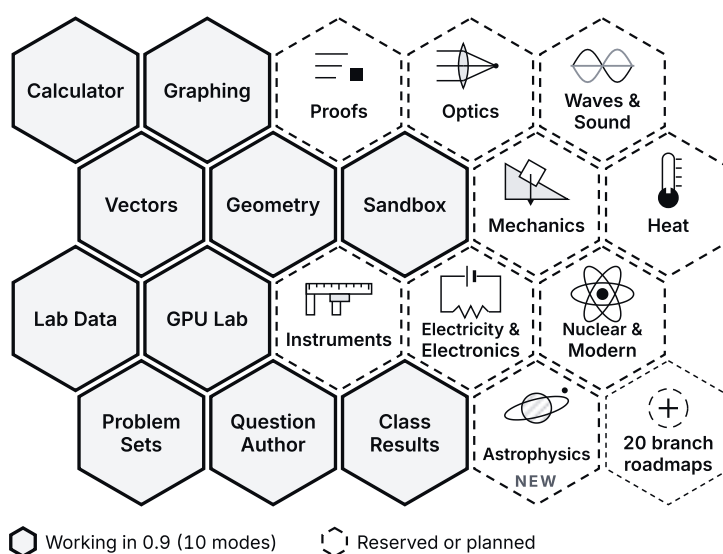
Heat · the gas laws, up to entropy [roadmap 21](#)

Electricity & Electronics · circuits, up to Maxwell's equations
[roadmap 22](#)

Nuclear & Modern · photons, up to quantum wells [roadmap 23](#)

Astrophysics · from the scale of things to cosmology
[Ideas 13–16](#)

Roadmaps 18 to 37 cover every branch of physics and maths. They are being written now.



VIRTUAL LAB AND INSTRUMENTS · SANDBOX PRO

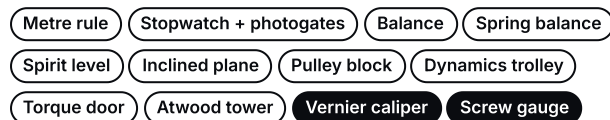
Measure it, record it, rewind it

Instruments on the real physics engine, a recorder that captures what really happens, and a Sandbox that can stop time.

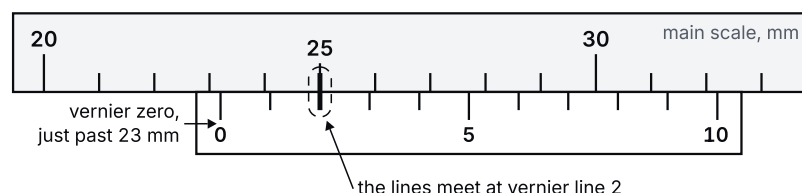
FUTURE IDEA 2 · THE INSTRUMENTS MODE

A lab that reads to the last division

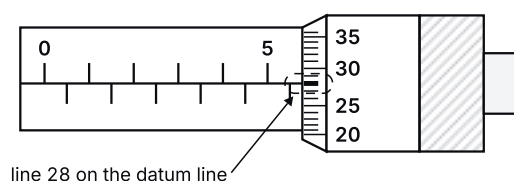
Twelve working instruments. Every reading carries its uncertainty, goes to Lab Data in one click, and is tested against the textbook formula.



Filled: added by the research, since they open almost every course. Six university instruments follow, from the oscilloscope to the Hall probe.



Reading = 23 mm + 2 × 0.1 mm = 23.2 mm ± 0.1 mm



5.5 mm + 28 × 0.01 mm = 5.58 mm ± 0.01 mm

FUTURE Read the scale yourself; PhysLab checks your reading, zero error included. Left: ten vernier lines span 9 mm, so it reads to 0.1 mm. Right: one turn of the thimble moves 0.5 mm over 50 lines, so it reads to 0.01 mm. Every instrument has a test: the Atwood tower's simulated 1.95 ± 0.04 m/s² must contain the formula's 1.962 m/s².

FUTURE IDEA 8 · DATA CAPTURE

Record often enough to see the truth

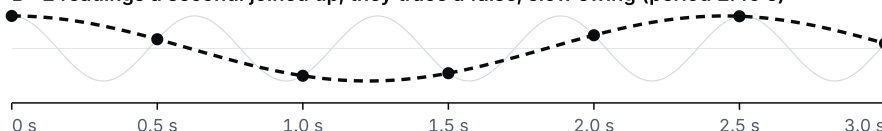
Three ways to capture, as school data loggers do: **continuous** at an exact rate from $t = 0$; **events**, one row per bounce, gate or swing; and **keep**, one row per press. It records x , y and z for several bodies at once.

If the rate is too slow for the motion, it says so in a sentence and shows why.

A · 10 readings a second: the dots follow the true swing (period 0.628 s)



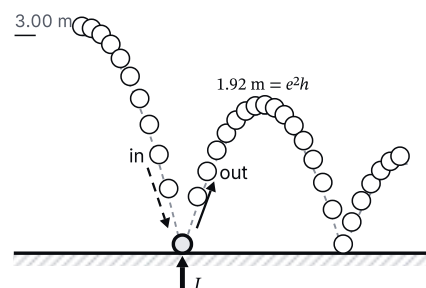
B · 2 readings a second: joined up, they trace a false, slow swing (period 2.45 s)



FUTURE Why a slow rate lies. A mass on a spring (2 kg on 200 N/m) swings 1.59 times a second, so it needs more than 3.2 readings a second (the Nyquist rule).

FUTURE IDEA 7 · SANDBOX PRO

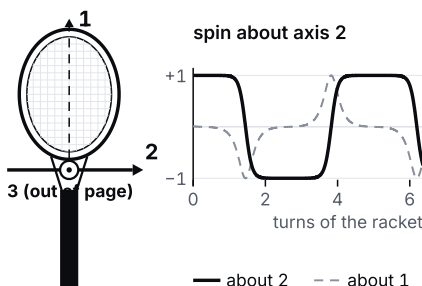
A Sandbox with a rewind button, and physics you can check



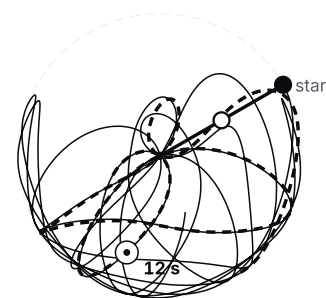
The bounce inspector. A 0.5 kg ball dropped from 3 m, stopped at the contact: in 7.65 m/s, out 6.13 m/s, so $e = 0.80$; the floor's push $J = 0.5 \text{ kg} \times 13.78 \text{ m/s} = 6.89 \text{ N s}$; 36 % of the energy ($1 - e^2$) becomes heat and sound.

FUTURE Bounce numbers from a real run of the physics engine; the flip and the twin runs from their equations of motion, solved with energy kept to better than one part in 10^9 .

A recorder keeps every physics step, so a run can be rewind, slowed down smoothly and stepped one frame at a time. Choose 60, 240 or 600 steps a second and watch the error shrink. Deeper rungs add the full inertia of a spinning body, chaos and a reference solver.



The tennis-racket flip. A spin about axis 1 or 3 is steady. About the middle axis, 2, the racket turns over again and again. Computed from Euler's equations (inertia 1 : 10 : 11).



Chaos, measured. Two double pendulums started 10^{-9} rad apart (solid, dashed) follow the same path for 12 s, then part. Drawn to scale from the equations of motion.

LAB DATA AND GRAPHS · SEEING IN 3-D

Data you can trust, graphs you can hear, solids you can print

FUTURE IDEA 9 · A MODERN LAB DATA TABLE

Build a column by tapping, not typing code

A grid that works by keyboard and touch, copies and pastes with Excel, and has one undo and redo. Worked-out columns are built from chips, and their units come for free.

1 $T^2 = \boxed{T}^2$ unit: s^2 (worked out)

worked out from chips				
#	l / m	t_{20} / s	T / s	T^2 / s^2
1	0.200	17.94	0.8970	0.8046
2	0.400	25.37	1.269	1.609
3	0.600	31.08	1.554	2.415
4	0.800	35.89	1.795	3.220
5	1.000	40.12	2.006	4.024

4 $\boxed{l} + \boxed{T}$ You cannot add metres to seconds.

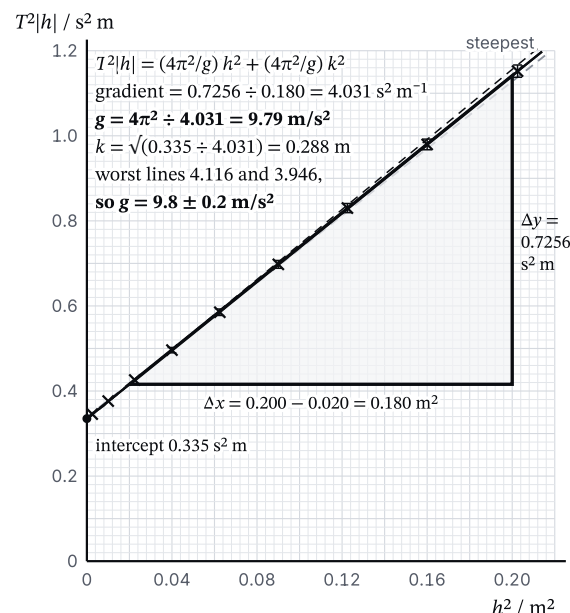
- Tap chips, constants and function buttons; the unit, s^2 , is worked out.
- Readings keep the digits typed: 0.200 stays 0.200.
- $25.37 \div 20 = 1.2685$ rounds to 1.269, on the decimal digits, as on paper.
- A sum of different units is refused in words.

FUTURE A pendulum table. Readings of $T = 2\pi\sqrt{l/g}$ with $g = 9.81$ m/s^2 ; "Graph these" hands them to Idea 5.

FUTURE IDEA 5 · THE GRAPH PAGE

Graphs drawn the way a careful student draws them

Graph paper, 1-2-5 scales, "quantity / unit" axes, error bars, the steepest and shallowest lines, and a big gradient triangle with its working written beside it. It exports a true-size PDF.

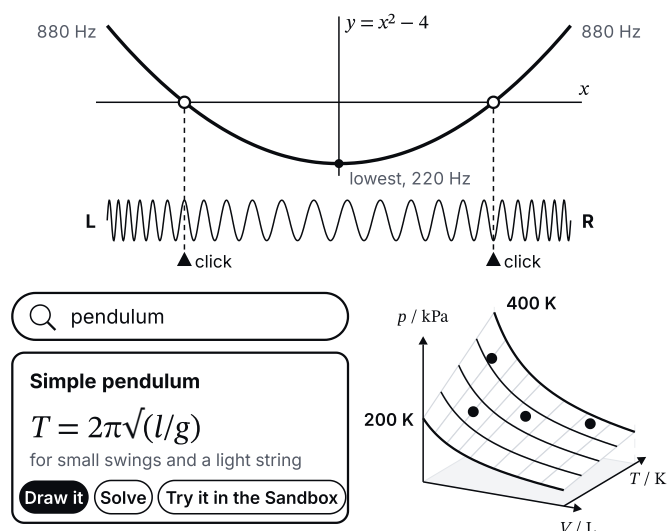


FUTURE The tester's own practical: a compound pendulum. A 1 m bar on 2 mm paper, drawn smaller than true size. Both of the template's methods give $g = 9.79 \text{ m/s}^2$ and $k = 28.8 \text{ cm}$. Illustrative readings.

FUTURE IDEA 3 · SOUND, FORMULAS, 3-D DATA

Hear a graph, find a formula, plot in 3-D

Any curve can be played as a tone whose pitch follows it. A searchable library of formulas opens each one drawn, solved or tried out. Three columns of Lab Data become points or a surface.

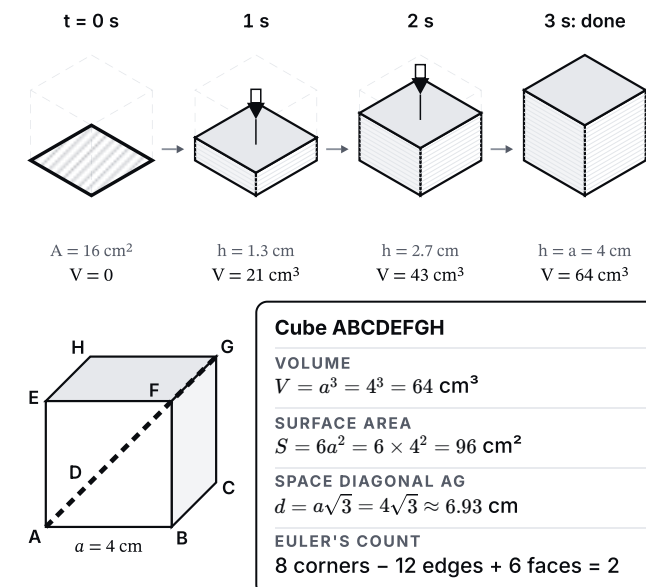


FUTURE Top: $y = x^2 - 4$, played left to right. The note falls to 220 Hz at the vertex, clicks at each root and moves from the left ear to the right. Pitch spans two octaves on a log scale, as hearing does. Right: $pV = nRT$ for 1 mol, with readings on it.

FUTURE IDEA 6 · SOLID GEOMETRY

Press U: a square becomes a cube

Any closed shape can be upgraded with one key: a square to a cube, a circle to a sphere. The solid prints itself layer by layer while the volume counts up, which is Cavalieri's principle at work.



FUTURE Exact first, then rounded. Every answer is checked a second way before it is shown. Solids can also be taken apart (a frustum is a cone minus a cone), unfolded into nets and cut by a plane.

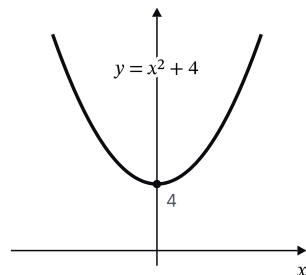
CALCULATOR · UNIVERSAL PROBLEM SOLVER AND CREATOR

One switch for complex numbers, and a solver that shows its chain

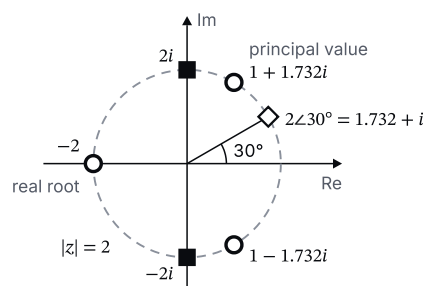
FUTURE IDEA 10 · REAL OR COMPLEX, AND A FULLER KEYPADFlip one switch, and every answer may use i

One switch, $\mathbb{R}$ or $\mathbb{C}$, beside DEG and RAD. Every place that solves, factorises or calculates reads it. In $\mathbb{R}$ an impossible answer is a plain sentence with a one-tap "Turn on $\mathbb{C}$ ". In $\mathbb{C}$, Factorise splits $x^2 + 4$ into $(x - 2i)(x + 2i)$ in the same job.

The keypad grows to 44 px keys with the Shift meaning printed on each, a letters layer, an equals sign for equations, and a complex group.

 $\mathbb{R}$ $\mathbb{C}$ 

It never meets the x axis:
no real roots.

 $\mathbb{R}$ $\mathbb{C}$ 

■ roots of $x^2 + 4 = 0$ ○ cube roots of -8 ◇ $2\angle 30^\circ$, typed

FUTURE The same questions in $\mathbb{R}$ and in $\mathbb{C}$.

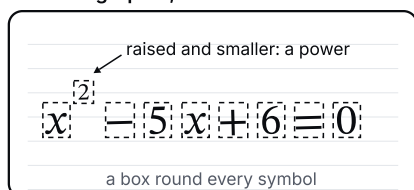
Every point lies on the circle $|z| = 2$: the roots of $x^2 + 4 = 0$, all three cube roots of -8 (the real root -2 beside the principal value $1 + 1.732i$), and $2\angle 30^\circ$, typed as it is written.

$\sin^{-1}$ e^x letters equations complex
sin ln x = i
44 px keys; the Shift meaning printed above each

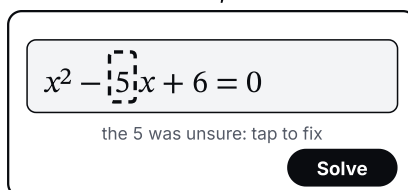
FUTURE IDEA 17 · PHOTO TO MATHS, BEING RESEARCHED

Photograph a sum; check it; get the exact answer

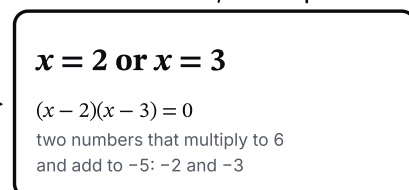
1 · Photograph it, or write it on the screen



2 · Check what it read, then solve



3 · The exact answer, with steps



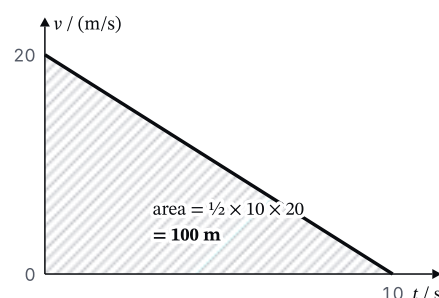
FUTURE No cloud, no account. Writing on the screen with a stylus, which is far easier to read, is also being studied. Equations only; word problems stay out.

FUTURE IDEA 11 · UNIVERSAL SOLVER AND CREATOR

Name what you know; it finds the chain of formulas

Enter the givens as chips with units and say what to find. A network of formulas finds a chain from the knowns to the unknown, checks every unit and draws the answer. With no AI it cannot read a word problem, so "comes to rest" is a chip for $v = 0$.

	KNOWN					
	u	v	a	t	s	
	20 m/s	0	-2 m/s ²	?	find	
$v^2 = u^2 + 2as$	●	●	●	○	○	① only s unknown: $s = 100$ m
$v = u + at$	●	●	●	○	○	② only t unknown: $t = 10$ s
$s = ut + \frac{1}{2}at^2$	●	○	●	●	○	② then, with $t: s = 100$ m
$s = \frac{1}{2}(u + v)t$	●	●	○	○	○	two unknowns: skip
● known ○ unknown One unknown in a row: that formula can be solved.						



The distance is the area under the line.

- ① **Chain found.** $0 = 20^2 + 2 \times (-2) \times s$, so $s = 400 \div 4 = 100$ m; the units are m²/s² on both sides.
- ② **Second route, as a check.** $t = 10$ s, then $s = ut + \frac{1}{2}at^2 = 200 - 100 = 100$ m. They agree.

One catalogue, shared with the formula library of Idea 3. About 150 formulas to start; every one has its units checked.

MOTION

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

ENERGY

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

CIRCUITS

$$V = IR$$

$$P = IV$$

GASES

$$pV = nRT$$

$$p_1V_1/T_1 = p_2V_2/T_2$$

WAVES

$$v = f\lambda$$

$$T = 1/f$$

FUTURE And a creator: any solved problem becomes a question with random numbers, whose later parts are marked with the student's own earlier answers, as real exams do.

ASTROPHYSICS 1 TO 4 · A NEW MODE

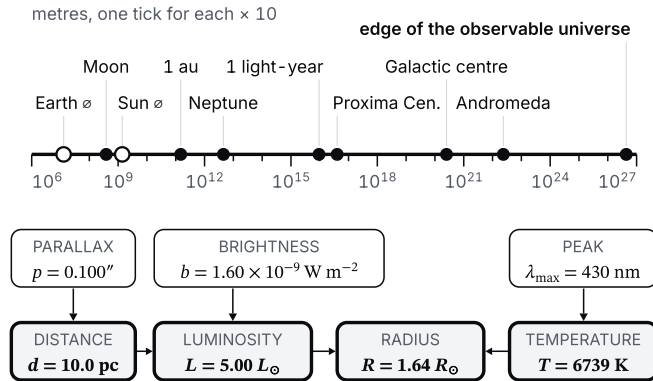
From a football Sun to the age of the universe

Students call it the hardest topic, so four ideas build a whole new mode for it. Every number on this page becomes a test.

FUTURE IDEA 13 · SCALE AND THE MEASURING CHAIN

Zoom from the Earth to the edge of everything

The mode opens on a Scale Ladder of real sizes (hollow) and distances (filled). Shrink it: with the Sun as a 22 cm football, the Earth is a 2.0 mm grain 23.7 m away. The measuring chain then runs from a parallax to a star's radius, either way.



FUTURE A slip caught: 430 typed in metres instead of nanometres gives a star of $1.1 \times 10^{27} \text{ m}$, bigger than the observable universe, so PhysLab asks you to check the unit.

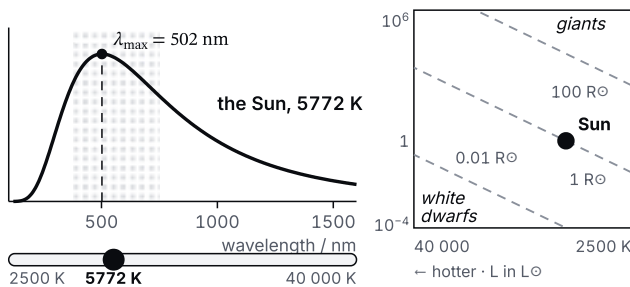
1.30 pc Proxima, from $0.768''$ **46.1** billion ly to the edge

6350 km Proxima, football scale

FUTURE IDEA 15 · THE STARS LAB

Drag the temperature; watch a star change

Four linked views: light, the HR diagram, inside a star, and its life. The slider drives a Planck spectrum, Wien's peak and the Stefan-Boltzmann power. Deeper rungs build model stars and find the white-dwarf limit.



FUTURE Left: the Sun, with the visible band dotted. Right: every dashed line is one radius, from $L = 4\pi R^2 \sigma T^4$. Research corrected the owner's notes: doubling a Sun-like star's mass makes it about 21 times brighter, and neutron stars top out near 2 to 2.5 solar masses.

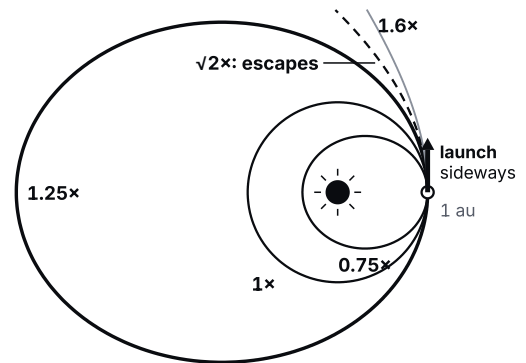
502 nm the Sun's peak **1.456 $M_{\odot}$** white-dwarf limit, $\mu_e = 2$

$3.828 \times 10^{26} \text{ W}$ the Sun

FUTURE IDEA 14 · THE GRAVITATION AND ORBITS LAB

Throw it fast enough and it never comes down

One Orbit Builder, five depths: Newton's cannon for a child; Kepler's laws and escape speed found by dragging; transfers, Lagrange points and tides; relativity at the top. It runs on its own accurate engine, not the Sandbox's.



FUTURE Drawn to scale from the computed orbits. Launch speeds are multiples of the circular speed at 1 au, 29.78 km/s. At 1.25x the path is an ellipse with $e = 0.5625$ and a period of 3.456 years; at $\sqrt{2}x$ the body escapes.

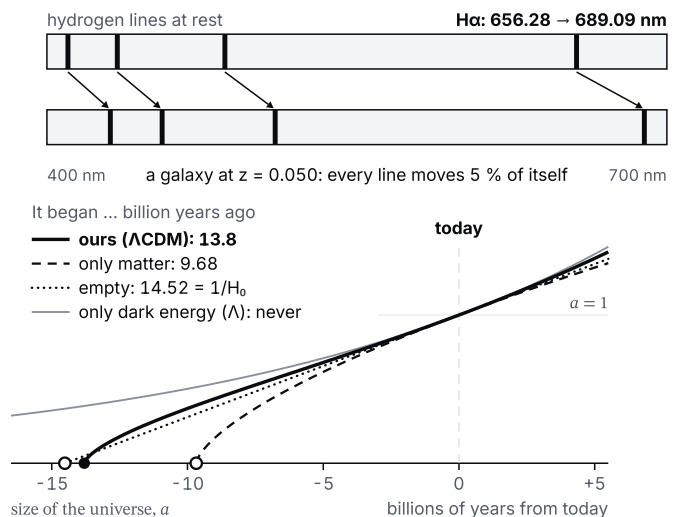
42 164 km geostationary radius **42.12 km/s** escape speed at 1 au

42.98'' Mercury, per century

FUTURE IDEA 16 · GALAXIES AND COSMOLOGY

Stretch the light; find the age of everything

A Redshift Lab turns a shifted line into a speed, and a Hubble diagram gives a first age, $1/H_0$. Go deeper for an exact cosmology calculator, rotation curves with dark matter, and fits to supernova data.



FUTURE Four universes with the same Hubble constant today began at different times, so $1/H_0$ is only an estimate of the age.

13.797 Gyr age, Planck 2018 **689.09 nm** $H\alpha$ at $z = 0.05$, in air

14.52 Gyr $1/H_0$

Also in these four ideas, each with its own known answers: Kepler's third law from the real planets · transfer orbits · Lagrange points · tides and the Roche limit · exoplanet transits · Mercury's precession · why hydrogen lines peak near 9 900 K · model stars · the CMB at 2.7 K · rotation curves and dark matter · colliding galaxies · lensing · the first helium.

FULL RIGOUR · THE 1.0 DESTINATION · THE SECOND BATCH

Proved right, shared freely, open to every learner

FUTURE IDEA 12 · RIGOROUS TESTING AND FULL ACCURACY

Every answer can say how we know it is right

IN 0.9 NOW Already in 0.9: one command runs all 2,879 tests, the type check and the lint before a release; the constants are CODATA 2022.

FUTURE Next: SymPy and mpmath as offline referees, thousands of generated cases, physics checked at two step sizes.

1 · First look



$3 \times 4 = 12 \checkmark$

the check, drawn as dots

2 · School

$2x + 3 = 13$
 $x = 5$
 $2 \times 5 + 3 = 13 \checkmark$

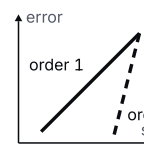
put the answer back

3 · College

$0.1 + 0.2$ is stored as
 0.30000000000000004
 and shown as 0.3

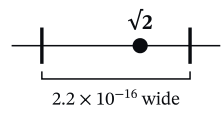
why exact fractions matter

4 · University



halve the step: $\div 2$ or $\div 16$

5 · Research



two neighbouring numbers

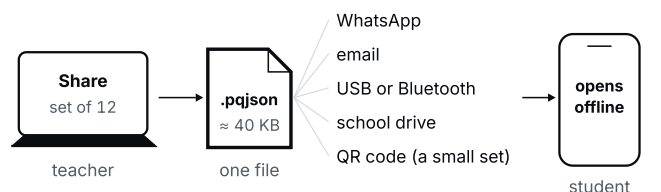
FUTURE The "How do we know?" card at five depths. The same check, from a picture a child can count to a proof that $\sqrt{2}$ lies between two neighbouring computer numbers.

FUTURE IDEA 4 · REDESIGN, SHARING, POLISH, PHONES

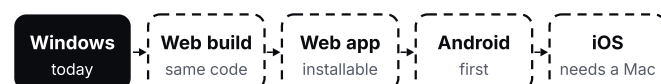
A 1.0 that strangers can trust, share and carry

A redesign to the standard of the best maths apps, built on the four themes already in place. Polish for people who have never met PhysLab: a first-run tour, a trusted installer and full documents. Question sets already travel as files in 0.9; a Share button will make any set, experiment or drawing one file.

FUTURE Phones after Windows. On a phone, heavy parts such as SymPy's steps load only when asked.



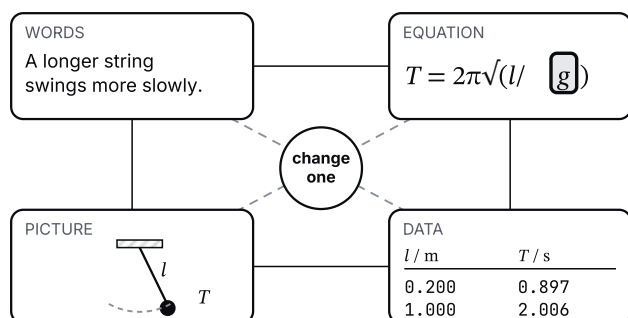
No server, no account, nothing uploaded. The file carries its licence.



Only about 12 small functions touch Windows, so a phone app wraps the same code: the same engine, files and themes. It is not a rewrite.

THE SECOND BATCH, QUEUED

Twenty branch roadmaps, and a learning layer across all of them



FUTURE The Four-Way Link. Words, equation, picture and data stay linked: change one and the other three follow. The words come from templates filled with live values, never from AI.

g · gravity near the ground
 unit m/s^2 · 9.81 on Earth, 1.62 on the Moon
 what if? drag it and watch the swing slow



FUTURE Living Equations: tap any symbol for its meaning, unit and typical values, with a what-if slider.

Every mode at full depth: mechanics up to Lagrange's $\frac{d}{dt} \partial_q L = \partial_q L$, electricity up to Maxwell's $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial_t \mathbf{E}$, and every branch of maths.

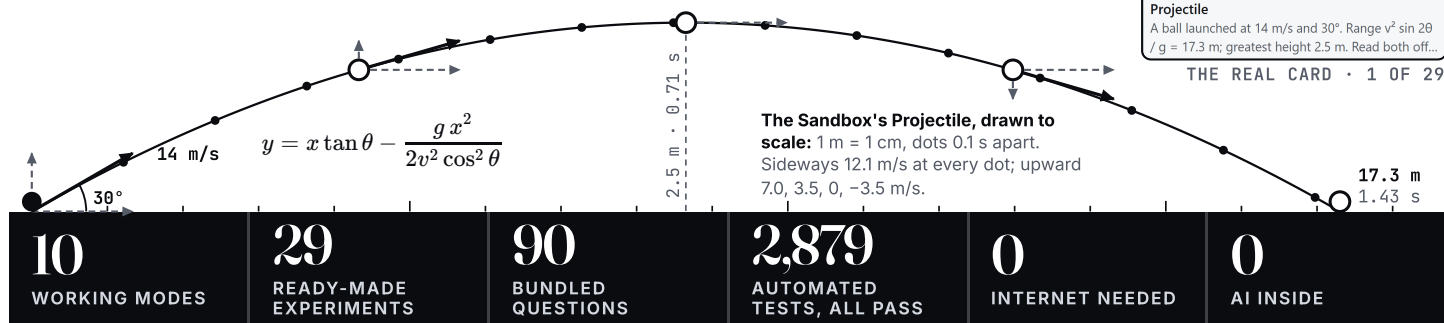
A classroom with no internet: on the school's own network the teacher sends a scene to every screen and collects the answers.

Runs on any laptop: quality tiers set by a quick automatic benchmark.

Open to everyone: screen readers, spoken maths, high contrast, dyslexia-friendly text, colour-blind-safe palettes, keyboard-only use, and languages such as Urdu, right to left.

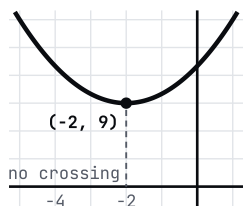
Whatever comes next keeps the rules PhysLab never breaks: no AI inside, fully offline · it feels like a calculator, never like programming · the answer comes first · every number carries its units · modes, never grade levels · anything solved can be seen.

Maths and physics you can see.



CALCULATOR • EXACT ALGEBRA 01

This parabola never touches the x-axis. So where are its roots?



ANSWER

$$x_1 = -2 + 3i$$

$$x_2 = -2 - 3i$$

✓ The two roots are conjugates — that always happens when the coefficients are real — and each one makes the equation zero exactly.

Tested. Typed $x^2 + 4x + 13 = 0$, pressed Work it out. The answer comes first, checked; the 7 steps wait until you ask.

1 Read off a, b, c

$$a = 1, \quad b = 4, \quad c = 13$$

2 Work out the discriminant

$$\Delta = (4)^2 - 4(1)(13) = -36$$

3 See what kind of root

$$\Delta < 0 \Rightarrow \text{two complex roots}$$

4 Write the root with i

$$\sqrt{-36} = \sqrt{36} \times (-1) = 6i$$

5 Use the quadratic formula

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2}$$

6 Read off the roots

$$x = -2 + 3i \text{ or } x = -2 - 3i$$

7 Check against the coefficients

$$x_1 + x_2 = -4 = -\frac{b}{a}, \quad x_1 x_2 = 13 = \frac{c}{a}$$

Look closer. The lowest point is $(-2, 9)$ and the roots are $-2 \pm \sqrt{9}i$.

LAB DATA 06

$$d = 4.92 \text{ tsq} + 0$$

$r^2 = 0.9997$ — the readings sit very close to this line.

$$\text{gradient} = 4.92 \pm 0.04 \text{ m/s}^2$$

From your error bars: the steepest line through them gives 4.98, the shallowest 4.9, and half the

Your readings, an honest $\pm$.

Five timed drops: gradient 4.92 $\pm$ 0.04 m/s², so $g = 2 \times \text{gradient}$.

HOW IT IS BUILT

Electron • React • three.js

Jolt + SymPy in WebAssembly

63,516 lines, **2,879** tests. Coded by Anthropic's state-of-the-art AI, **Claude Opus 5** and **Opus 5.5**, directed by a Class 11 student. No AI inside the app.

PROBLEM SETS 07

$$F_x = 33.10 \text{ N}$$

That is $F \sin \theta$. The x-component uses cos, the y-component uses sin.

$$F_y = 43.92 \text{ N}$$

That is $F \cos \theta$. The y-component uses sin.

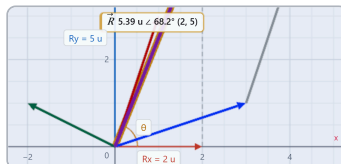
It names your mistake.

Fresh numbers every time, one hint at a time. 90 questions bundled.

11 DAYS, FIRST CODE TO 0.9

- 14 Sep First code
- 16 Sep Problem Sets • 17 Lab Data
- 18 Sep Sandbox on Jolt Physics
- 19 Sep Exact working, step by step
- 23 Sep 0.7: Geometry Lego, Numbas
- 25 Sep 0.9 preview: 10 modes

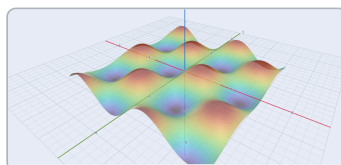
VECTORS 02



Add them. See the proof.

$R = 2i + 5j$: 5.39 u at 68.2°, drawn head to tail with its parts.

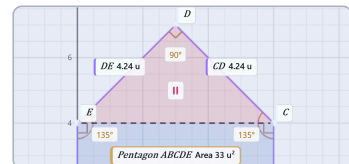
GRAPHING 04



Curves, regions, surfaces.

$z = \sin x \cos y$ in 3-D, $r = 2 \cos 3\theta$ on polar paper, $y > x^2 - 4$ shaded.

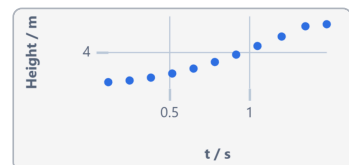
GEOMETRY 03



It knows the shape.

House = $24 u^2 + 9 u^2 = 33 u^2$. Lego pieces turn, flip and fuse.

SANDBOX 05



Real physics, by the book.

Atwood machine on Jolt Physics: the 1 kg rises at 3.27 m/s².

FOR TEACHERS 08

Open a question file...

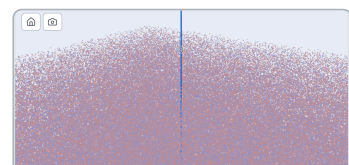
Save the set for Problem Sets

Save as a Numbas exam

Question Author, Class Results.

Every student gets new numbers. Results stay on the teacher's own PC.

GPU LAB 09



Millions of charged particles.

$F = q(E + v \times B)$ for each one. Here, 1,000,000 on the graphics card.

COMING NEXT • FUTURE

- Astrophysics:** scale, orbits, stars, galaxies.
- Instruments:** every reading with its $\pm$.
- 3-D geometry:** solids, nets, $V = E + F = 2$.
- Phones:** Android and iOS.

TRY IT AT THE STALL

- Type $x^2 + 4x + 13 = 0$. Guess the roots, then check.
- Sandbox ► **Projectile** ► Play. Find the 17.3 m.
- Geometry: open the house example. Press **Decompose**.

Free for the next student who needs it.